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In situ encapsulated Co/MnO_x nanoparticles inside quasi-MOF-74 for the higher alcohols synthesis from syngas

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ABSTRACT

Selective conversion of syngas (CO/H_2) to higher alcohols ($C_{2+}OH$) is of great interest but presents a significant challenge in keeping an appropriate balance between the carbon chain growth and CO insertion to achieve a high $C_{2+}OH$ selectivity. Herein we found that a core-shell Co/MnO_x@quasi-MOF-74 catalyst can be easily constructed through controlled deligandation of a bimetallic CoMn-MOF-74 by partial pyrolysis strategy. The as-obtained Co/MnO_x@quasi-MOF-74 catalyst produces three types of synergistic active sites (Co°, coordinatively unsaturated sites (CUSs) of Co^{2+} and Co_2C) that collaborate with each other to enhance $C_{2+}OH$ formation during the reaction. The Co° nanoparticles within the framework of quasi-MOF-74 enable CO dissociation and significant CH_x-CH_y coupling to occur while the uniformly distributed CUSs of Co^{2+} working with Co_2C strengthen CO insertion process, leading an outstanding catalytic performance in the process of CO hydrogenation. The total selectivity of alcohols (ROH) reached 48.7 wt%, where of 93.2 wt% can be $C_{2+}OH$, and very low CH_4 and indetectable CO_2 were produced at 200 °C, 3.0 MPa (CO/H_2 = 1/2) and a gaseous hourly space velocity (GHSV) of 4500 mL g^{-1} h^{-1}, reaching the catalytic performance comparable to that of the optimal level of multifunctional catalyst operated at much higher pressure (6.0 MPa). This work highlights the potential of using MOF-derived quasi-MOF materials as a tunable platform to explore highly efficient catalysts for syngas conversion.

1. Introduction

Catalytic conversion of syngas (CO/H_2) derived from natural gas reforming or coal/biomass gasification into clean fuels and value-added chemicals is of great interest for its potential role in the sustainable carbon cycle [1–3]. Higher alcohols synthesis (HAS) from hydrogenation of CO is economically attractive because of their potential use as fuel or fuel additives, and other platform molecules. However, this process is challenging with respect to the requirement of C—C coupling while retaining some of the C—O bonds [1,3]. The low yield and selectivity of $C_{2+}OH$ in HAS, to a great extent, hinder its industrial application [4]. Typically, four types of catalysts (1) (Co-, Fe-, Ni-based) modified Fischer-Tropsch synthesis (FTS) catalysts [5–13]; (2) Rh-based catalysts [14,15]; (3) MoS_2- and Mo_2C-based catalysts [16,17]; and (4) (Cu-based) modified methanol synthesis catalysts were used for HAS [18–21]. Among these catalytic systems, Co-based FTS catalysts are always preferred over others due to their comparatively C_{2+} alcohols selectivity, low cost and good stability [3]. Nevertheless, it is well-known that, for the Co-based FTS catalysts, CO was found to be dissociatively adsorbed on top of Co^0 sites, which would facilitate the CH_x formation in HAS process [1,3,5,14,22]. Thus, some of other elements, such as Cu, Fe, Ni and Mn, are generally used to provide some of CO nondissociative sites to facilitate the formation of $C_{2+}OH$ [1,4]. In other words, to improve the selectivity of $C_{2+}OH$, an appropriate balance between the carbon chain growth and CO nondissociative insertion needs to be kept delicately [1–3].

Metal-organic frameworks (MOFs), emerged as a new class of crystalline porous organic-inorganic hybrid materials, provide a tunable platform for the design of various functional materials due to their unique crystalline structure, atomic metal dispersion, controllable

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porosity and textural properties [23–25]. Using MOFs as precursors for the synthesis of FTS catalysts seems to be very promising [26]. Several studies showed that MOFs-based or derived catalysts resulted in an unrivalled performance in FTS process to produce long chain hydrocarbons [27–35]. However, there is almost no research into the MOFs-based catalysts for the HAS because of its complex reaction networks and strict operation condition. For MOF materials, structural stability and suitable active sites are essential for achieving a high $C_{2+}OH$ selectivity in HAS process. In general, compared with pristine MOFs, MOF-derived materials obtained by direct pyrolysis of MOF crystals are more favored in FTS because of their thermal stability [34,35]. But these MOF-derived materials usually have a lower surface area and collapsed skeletal structure after high temperature decomposition, resulting in a large amount of active metal sites being embedded into the bulk phase and coated with graphite carbon, which prevents the reactants from closing to the reactive centers. Thus, we envisioned that a transitional MOF material (quasi-MOF) inherited the structural skeleton, porosity of pristine MOF and stability of MOF derivatives could be created through strictly controlled deligandation [36–38]. Based on this assumption, we chose bimetallic CoMn-MOF-74 as a precursor to synthesize such a quasi-MOF derivative. And the following factors are mainly considered: (1) The hydroxyl and carboxyl oxygens of 2,5-dihydroxyterephthalic acid (H_4DOT) are fully coordinated with divalent metals, which makes the framework of MOF-74 very stable. (2) The extra high density of metal sites can provide plenty of reactive centers after controlled thermal treatment. (3) MOF-74 possesses a large, one-dimensional hexagonal pore system of 1.1–1.2 nm in diameter, which is large enough for the diffusion of HAS products timely. (4) Different bonding strengths between the divalent transition metals and ligands result in a sequential degradation of Co and Mn during the process of controlled deligandation, which can form highly active heteroparticles [39].

Herein, we constructed Co/MnO_x nanoparticles in situ within the formwork of MOF-74 to form a Co/MnO_x@quasi-MOF-74 core-shell catalyst by controlled deligandation of bimetallic CoMn-MOF-74. In this construct, the porosity and skeleton structure of pristine CoMn-MOF-74 were largely retained, ensuring the reactants accessible to the active sites as well as to diffuse in a timely and effective manner. More importantly, three types of synergistic active sites (Co°, coordinatively unsaturated sites (CUSs) of Co^{2+} and Co_2C) were created in HAS, where metallic Co° inside the pore of quasi-MOF-74 worked for CO dissociation and chain growth for C_{2+} alkyl chain while CUSs of Co^{2+} and Co_2C for CO insertion to generate $C_{2+}OH$. As expected, due to the cooperative nature of these three reactive centers, the as-prepared Co/MnO_x@quasi-MOF-74 exhibited an outstanding catalytic performance during the reaction. The selectivity of alcohols (ROH) reached 48.7 wt% with a CTY (activity per gram of catalyst) of 117.8 $mg_{co}\,g_{cat}^{-1}\,h^{-1}$, whereof up to 93.1 wt% can be $C_{2+}OH$ at 200 °C, 3.0 MPa (CO/H_2 = 1/2), and a GHSV of 4500 mL $g^{-1}\,h^{-1}$. Meanwhile, sum selectivity of ROH and olefins reached 69.1 wt% and very little CO_2 was detected in the products. To the best of our knowledge, this is the first report of a MOF-based heterogeneous catalyst with multi active sites was used to synergistically catalyze the conversion of syngas to higher alcohols.

2. Experimental

2.1. Catalysts synthesis

The Co/MnO_x@quasi-MOF-74 and Co/MnO_x@C-X catalysts were obtained by controlled deligandation of a bimetallic CoMn-MOF-74 precursor (Fig. 1). Firstly, we synthesized a bimetallic CoMn-MOF-74 by the hydrothermal method. Typically, a solid mixture of Co$(NO_3)_2$·6H_2O (0.52 g, 1.787 mmol), $MnCl_2$·4H_2O (0.26 g, 1.314 mmol) and H_4DOT (0.20 g, 1.009 mmol) was added to a mixture of DMF/ethanol/water = 55.0/3.5/3.5 mL in a 100 mL Teflon-lined autoclave, and stirred for 30 min until became homogeneous, following by a 24.0 h thermal treatment at 135 °C. After cooling to room temperature, the resulting crystals were soaked and washed with fresh MeOH solvent for 6 times over a 3 days period, until the supernatant was transparent. Then it was activated to remove all solvent under vacuum at 40 °C for 12.0 h, yielding the deep-purple crystalline porous material. Secondly, the powder of CoMn-MOF-74 (about 1.0 g) was located horizontally in a ceramic fiber oven and placed in a quartz tubular reactor (approximately L = 1.0 m × ID = 4.5 cm). The sample was subjected to heat treatment with a heating rate of 1.0 °C min^{-1} and dwell time of 4.0 h in nitrogen atmosphere (around 50 mL min^{-1}) varied at different temperature of 400, 500, 600, 700 and 800 °C. The synthesized catalysts were denoted herein as Co/MnO_x@C-X with X representing the pyrolysis temperature in °C. The Co/MnO_x@C-400, namely, is Co/MnO_x@quasi-MOF-74, while Co/MnO_x@C-500, 600, 700 and 800 are the Co/MnO_x@C catalysts. The Co@quasi-MOF-74 and MnO_x@quasi-MOF-74 were obtained by directly pyrolyzed Co-MOF-74 and Mn-MOF-74 at 400 °C with the same procedure as Co/MnO_x@quasi-MOF-74. The detailed synthetic process of Co-MOF-74 and Mn-MOF-74 were shown in Supporting Information.

Hydrothermal synthesis of bimetallic CoMn-MOF-74. ② Controlled deligandation of CoMn-MOF-74 precursor by the controlled thermal treatment strategy to form a Co/MnO_x@quasi-MOF-74 core-shell catalyst. The resulting composite not only retained the framework of pristine MOF-74, but also confined Co/MnO_x within its framework, which is an excellent catalyst for the HAS. ③ Direct pyrolysis of CoMn-MOF-74 precursor at higher temperature (> 500 °C) to form Co/MnO_x@C core-shell catalysts. The frameworks of these catalysts are completely collapsed, which are inefficient in HAS.

2.2. Catalyst characterization

Powder X-ray diffraction (XRD) patterns were collected on a Rigaku MiniFlex 600 diffractometer with Cu Kα radiation (λ = 1.5418 Å). Metal contents were determined by inductively coupled plasma optical emission spectrometry (ICP-OES) analysis (Perkin-Elmer 3300DV). Carbon, hydrogen, and nitrogen elemental microanalyses (EA) were performed on a Vario EL cube CHONS elemental analyzer. The morphology of the catalysts was examined by HRTEM-EDX (Philips Tecnai G2 F20 microscope) and SEM-EDS (Zeiss Gemini Ultra-55). The XPS data were recorded on ESCALAB 250Xi X-ray photoelectron spectrometer. The Raman spectrum was obtained on a RTS-HiR-AM Raman spectrometer with laser excitation at 532 nm wavelength detector. Fourier transform infrared spectra (FT-IR) were performed on a Bruker TENSOR 37. Nitrogen adsorption/desorption isotherms were obtained with a Micrometrics ASAP 2460 volumetric gas adsorption analyzer. The samples were degassed at 150 °C in vacuum for 10.0 h before tests to confirm the entire removal of overall guest molecules or adsorbed moisture. The specific surface area was calculated using Brunauer-Emmett-Teller (BET) method. Mesoporous surface area, pore volumes, and mean pore diameters of mesopores were evaluated by Barrett-Joyner-Halenda (BJH) method from the adsorption branches of isotherms. Temperature programmed reduction with H_2 (H_2-TPR) was performed by using BELCAT II (Microtrac BEL) with a cryo apparatus. The sample was mounted in a cell and then heated at a ramping rate of 10 °C min^{-1} under a flow of 5% H_2/Ar (20 mL min^{-1}). Temperature programmed desorption (TPD) test was performed on a chemisorption analyzer (Autochem 2950HP) from Micromeritics. Before test, the catalyst was heated at 250 °C for 2.0 h in 50 mL min^{-1} in N_2, and then reduced by 50 mL min^{-1} pure H_2 at 300 °C for 2.0 h. After the reduction, the sample was purged with 50 mL min^{-1} He for 2.0 h, and the temperature was cooled to 50 °C, CO and H_2 adsorption began with 50 mL min^{-1} 10 % CO-90 % He and 10 % H_2-90 % He for 2.0 h, respectively. Finally, the desorption proceeded from 50 to 300 °C at 10 °C min^{-1}.

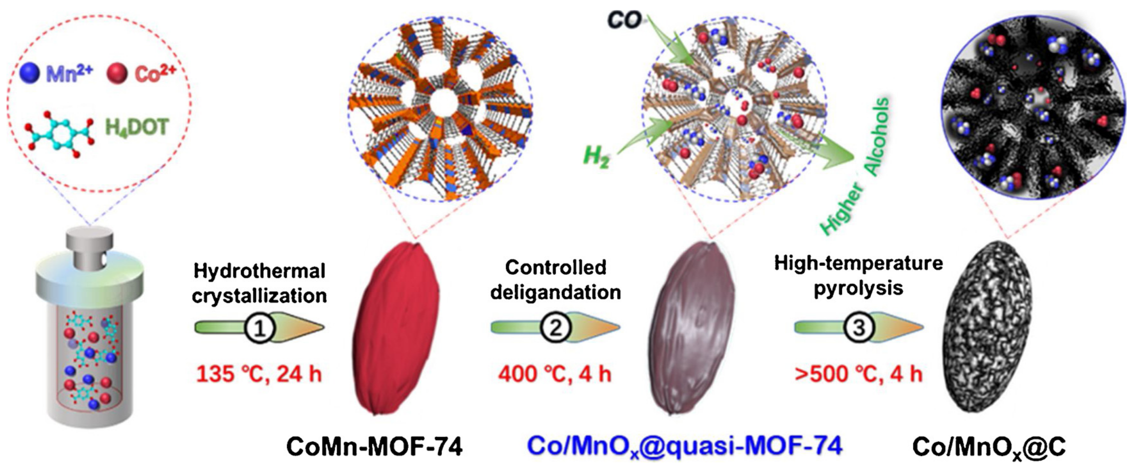

Fig. 1. Schematic illustration of the syntheses of Co/MnO_x@quasi-MOF-74 and Co/MnO_x@C catalysts. ①.

2.2.1. In situ infrared spectroscopy

In situ diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) measurement was performed on a BRUKER FTIR spectrometer (Tensor II) with a liquid nitrogen cooled MCT detector. Prior to measurement, the catalyst was pretreated under N_2 at 250 °C for 30 min to remove any moisture and adsorbed gases, and then was in situ reduced at 300 °C under a H_2 atmosphere for 2.0 h, and finally it was flushed by pure N_2 for 1.0 h with the temperature cooling to 30, 230 or 250 °C. Then, the background spectrum (64 scans) with a resolution of 4 cm^{-1} was recorded. Subsequently, CO was introduced to the cell at specific temperature, which was held for 1.0 h for its full adsorption onto the sample. The in situ reaction tests were performed after the switch of the H_2 and CO flow with a molar ratio of 2:1 at a total flow of 30 mL min^{-1}.

2.2.2. Semi-in situ X-ray photoelectron spectroscopy

Semi-in situ XPS was performed at the Escalab 250Xi (XPS) measurements. The setup consisted of a special reaction chamber and an XPS analysis chamber. The catalysts were firstly pretreated in reaction chamber, and then were transferred to the analysis chamber for test without exposure to air. The pressure in the reaction chamber was controlled by a pressure controller to be 1.0 bar, and the gas flow was controlled by mass flow controllers. In this study, the catalysts were pretreated with N_2 at 250 °C for 2.0 h, H_2 at 300 °C for 2.0 h, and syngas (H_2/CO = 2) at 230 °C for 2.0 h for test, respectively. The binding energies in all the spectra were calibrated according to C 1s peak at 284.6 eV. We also referred the XPS binding energies to the NIST XPS database (https://srdata.nist.gov/xps/main_search_menu.aspx).

2.2.3. In situ Raman spectroscopy

For in situ Raman spectroscopy measurement, the sample was firstly dried in Ar (20 mL min^{-1}) at 200 °C for 2.0 h. Because much more amount of sample, which is often thousands or hundreds times that for XPS and IR characterization, is required for the in situ Raman spectroscopy experiment. To enable the Co^{2+} that resulted from part of Co° oxidization in air during sample packaging and testing can be completely reduced, we chose a slightly higher temperature (350 °C) to prereduce the dried samples in a H_2 (30 mL min^{-1}) for 2.0 h before measurement. After H_2 pretreatment, it was purged with Ar (20 mL min^{-1}) at 230 °C for 0.5 h. Subsequently, the sample was exposed to a H_2/CO/N_2 mixture flow (H_2/CO/N_2 = 63.5/31.5/5.0, 30 mL min^{-1}).

2.3. Catalytic tests

Syngas conversion was performed in a continuous flow fixed-bed stainless steel reactor (inner diameter, 6 mm) designed by Beijing TERCH Co., Ltd. The schematic of the setup apparatus is shown in Fig. S1. Typically, 0.40 g catalyst uniformly mixed with 1.00 g quartz sand (35 − 60 mesh) loaded in the constant temperature zone of middle reactor, and the detailed filling method thereof is shown in Fig. S2. The device was tested for leak using soap bubbles prior to the start of the reduction and reaction. After leak test, the catalyst was in situ pretreated with a high purity hydrogen at the atmospheric pressure, 300 °C for 2.0 h. After the pretreatment, the temperature in the constant temperature zone of the reactor was cooled to 230 °C, and the back pressure valve was adjusted to slowly increase the internal pressure of the reaction tube. When the pressure was raised to 3.0 MPa, the pure syngas (H_2/CO/N_2 = 63.5/31.5/5.0, 5.0 vol% of N_2 as the internal standard) was fed into the reactor with a GHSV of 4500 mL g^{-1} h^{-1}. Reaction was carried out under conditions of H_2/CO = 2, 3.0 MPa, 230 °C and 4500 mL g^{-1} h^{-1} unless otherwise stated. Data were collected after at least 24.0 h on stream until the reaction reached a stationary state. The mass balance, carbon balance and oxygen balance were kept between 90 % and 110 %. As shown in Fig. S1, a hot trap (120 °C) and a cold trap (0 °C) were employed to capture the liquid product in the effluent. To avoid possible condensation of the reaction products, the temperature of the whole pipelines from reactor to cold trap kept at about 150 °C during the catalytic tests. The gas products, after passing through the cold trap, were monitored by online gas chromatography with GC-6890, which was equipped with a thermal conductivity detector (TCD, a TDX-01 packed column was used to separate H_2, N_2, CO, CH_4, and CO_2) and a flame ionization detector (FID, equipped with a Al_2O_3/KCl capillary column to separate C_1-C_7 hydrocarbons). Besides, CH_4 was taken as a reference bridge between FID and TCD. The liquid phase product consists of three parts, an aqueous phase, an oil phase and a wax phase (Fig. S3, obtained from hot trap), wherein the aqueous phase was the most abundant, followed by the oil phase, accompanied by a trace amount of the wax phase. The details of gas chromatography analysis are illustrated in Figs. S4–S9.

The CO conversion (X_{CO}) is calculated by an internal normalization method and carbon balance method.

$$\chi_{CO} = \left(1 - \frac{A_{CO,\,out} \bullet A_{N_2,\,in}}{A_{CO,\,in} \bullet A_{N_2,\,out}}\right) \bullet 100\% \tag{1}$$

Where $A_{CO,in}$ and $A_{N2,in}$ are the peak areas of the corresponding feed gases (H_2/CO/N_2 mixture with a volume ratio of 63.5 %H_2/31.5 %CO/5.0 %N_2) in the TCD chromatographs before reaction. $A_{CO,out}$ and $A_{N2,out}$ are the peak areas of the exit gas composition during catalysis.

The selectivity of CO_2, individual hydrocarbon (RH) or ROH product S_i was calculated according to the following Eqs. (2–3):

$$S_{CO_2} = \frac{CO_{2,\,out}}{CO_{in} - CO_{out}} \times 100\% \tag{2}$$

$$S_{C_i}(i = 1,2,\ 3,4) = \frac{\sum_{k=1}^{4} nC_i}{CO_{in} - CO_{out}} \times 100 \tag{3}$$

$$S_{C_{5+}} = 100 - S_{CO_2} - \sum_{i=1}^{4} S_{C_i} \tag{4}$$

Where $CO_{2,out}$, CO_{out} and CO_{in} are the moles of CO_2 and CO at the outlet and CO at the inlet, respectively.

The liquid products are mainly RH and ROH, and the selectivity of a product is calculated as follows:

$$RH/ROH = \frac{Weight\ of\ RH/ROH}{Weight\ of\ all\ of\ products} \times 100\% \tag{5}$$

The chain growth probability (α) was calculated according to the following eq:

$$\ln(\frac{W_{n_i}}{n}) = n\ln\alpha + \ln\frac{(1-\alpha)^2}{\alpha} \tag{6}$$

Where n_i represents the carbon number of a product i (ROH or RH), and W_{ni} represents the mass fraction of the product i.

3. Results and discussion

3.1. Catalysts syntheses and structural analysis

The bimetallic CoMn-MOF-74 was synthesized via hydrothermal reaction of Co^{2+}, Mn^{2+} metal salts with H_4DOT in the presence of dimethylformamide/ethanol/water at 135 °C for 24.0 h (Fig. 1). The morphology of the as-prepared CoMn-MOF-74 was investigated using SEM. As seen from Fig. S10, SEM images present a rugby ball shape with a size of few micrometer scale. The crystalline structures of samples were examined by XRD. Fig. 2a shows the XRD patterns of CoMn-MOF-74 are almost identical to the simulated MOF-74 (Fig. S11), indicating a high phase purity [40]. Then, we used the controlled thermal treatment strategy to fabricate Co/MnO_x in situ within the framework of MOF-74.

The SEM images showed that all the samples obtained at different temperatures still maintained the morphology of the pristine CoMn-MOF-74 (Fig. S10), but the surface of the MOFs became rough and ultrafine nanoparticles started to emerge from the MOF substrates with the increase of calcined temperature (Fig. S12), which indicated the crystallinity had gradually damaged. The XRD pattern of Co/MnOx@quasi-MOF-74 is almost identical to that of as-synthesized CoMn-MOF-74 (Fig. 2a), and there is no any diffraction peaks attributed to Co/MnO_x nanoparticles because of their very small particles size and low diffraction peak intensity, suggesting that MOF-74 framework largely retained up to 400 °C. However, for the completely decomposed samples, namely Co/MnO_x@C-500, 600, 700, and 800, all of the characteristic peaks of MOF-74 disappeared completely (Fig. 2a and S11). Meanwhile, a broad characteristic diffraction peak at about 2θ = 43.91° corresponding to the (111) lattice plane of carbon (JCPDS No. 06-0675), and well-defined diffraction for the metallic Co (JCPDS No. 15-0806) and MnO (JCPDS No. 07-0230) were becoming more evident with the pyrolysis temperature increasing. The results are implying that MOF-74 nanocrystals had completely decomposed and new porous carbon materials generated after the calcination over 500 °C. The obtained results were also confirmed by FT-IR. As shown in Fig. 2b, after 400 °C calcination, no new peak appeared or any existing peaks ascribed to MOF-74 disappeared suggests that the framework of MOF-74

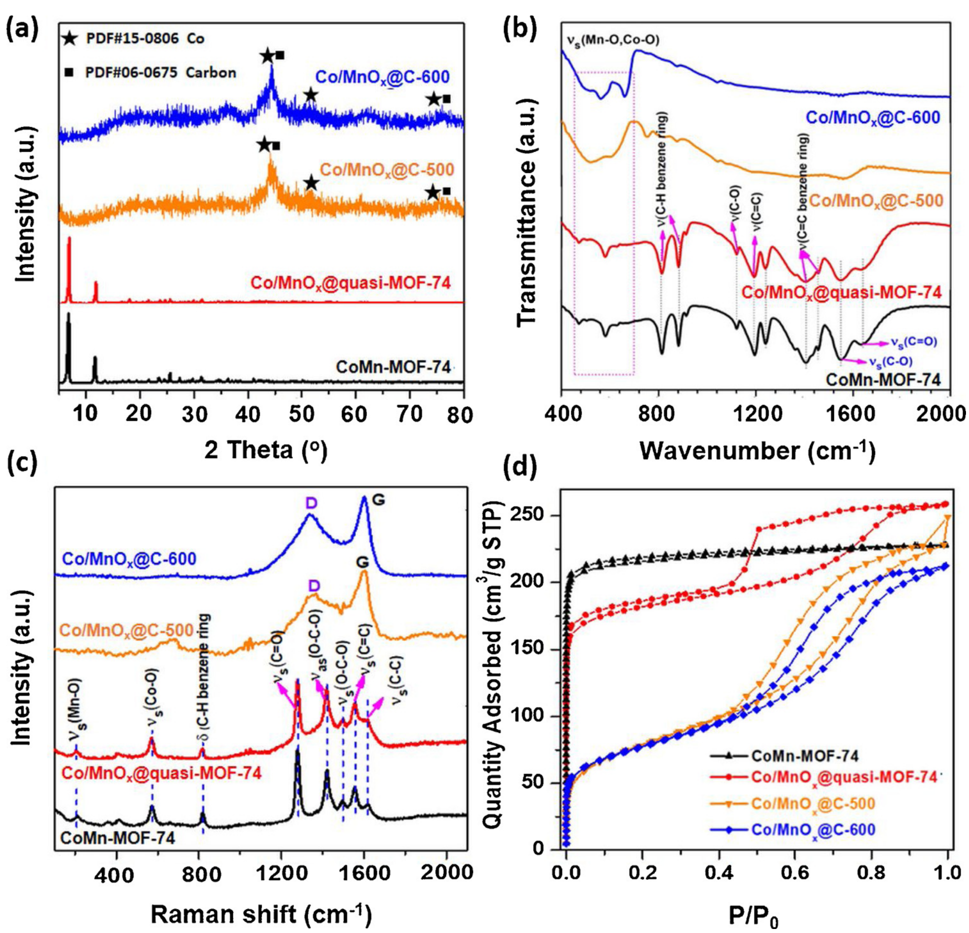

Fig. 2. Structural characterization of the CoMn-MOF-74, Co/MnO_x@quasi-MOF-74, Co/MnO_x@C-500, and Co/MnO_x@C-600 catalysts. (a) XRD patterns, (b) FT-IR spectra, (c) Raman spectra, and (d) N_2 sorption isotherms.

remained intact. But, the intensities of the FT-IR peaks at 1455 and 1400 cm^{-1} attributed to ν(C=C) of mononuclear aromatic hydrocarbons [40], which are the benzene ring skeleton vibration, slightly decreased, indicating the partial deligandation of the MOF-74 had occurred. When the annealing temperature raising up to 500 °C, the valuable peaks for carboxylate groups almost disappeared, which should be a result of complete carbonization of the organic moieties. Additionally, two new broad bands appeared at 700−500 cm^{-1}, which could be ascribed to the Co−O or Mn−O stretching vibration peak [40], indicating the formation of MnO_x or CoO_x at higher temperature. In agreement with the FT-IR results, the Raman spectra further verified the partial deligandation of MOF-74 at 400 °C and complete decomposition beyond 500 °C for 4.0 h. The vibration peaks of Co/MnOx@quasi-MOF-74 at 820, 1276, 1420 and 1494 cm^{-1} corresponding to C−H bending of the benzene ring, the C=O stretching, the O−C−O symmetric and asymmetric stretching of the carboxylate group of the organic linker respectively [41], remained, and basically the same as pristine CoMn-MOF-74 (Fig. 2c). Upon raising the calcination temperature to 500 °C, the vibration bands for organic linker almost disappeared, while the coexistence of D band (at around 1328 cm^{-1}) and G band (at around 1580 cm^{-1}) were observed, indicating the formation of disordered/defect graphitic carbon [42], in consistent with the results of XRD characterization. N_2 adsorption/desorption measurements showed that the Co/MnO_x@quasi-MOF-74 (552.71 $m^2\,g^{-1}$) retained 86.3 % of the Brunauer-Emmett-Teller (BET) specific surface area from CoMn-MOF-74 (640.48 $m^2\,g^{-1}$), while the Co/MnO_x@C-600 (202.56 $m^2\,g^{-1}$) only retained 31.64 %. Meanwhile, most of the micropores BET surface area and pore volume of MOF-74 remained after partial deligandation (Table S1, Fig. S13), indicating that Co/MnO_x@quasi-MOF-74 retained the MOF porosity to a great extent, in agreement with the XRD, FT-IR and Raman results.

The electron microscopy was used to reveal the nanostructure of these MOF-74 derived samples. As displayed in the high angle annular bright-field (HAABF) and dark-field (HAADF) images (Fig. 3d, e and S14a, b) from scanning transmission electron microscopy (STEM) and TEM (Fig. 3g, S14d, e) images, numerous highly-dispersed nanoparticles with an average diameter of 8.3 nm (Fig. S14f) were emerged in Co/MnO_x@quasi-MOF-74. The corresponding EDS elemental mapping revealed that C, O, and Mn elements were homogeneously distributed over the bulk phase of Co/MnO_x@quasi-MOF-74 (Fig. 3h and S15), while Co element not only distributed throughout support but also perfectly matched that of nanoparticles, suggesting that Co species participate in the composition of both the framework and nanoparticles. The high-resolution HAABF-STEM images (Fig. 3f and S16b) of the nanoparticles in Co/MnO_x@quasi-MOF-74 reveal that they possessed several distinct crystal structures. As can be seen from the enlarged images (Fig. 3i, j, m and n, which were selected from the areas marked by the rectangles in Fig. 3f), the clear lattice fringe with a D-spacing of 0.205 nm is corresponding to the (111) plane of Co, and the lattice spacings of 0.257, 0.222, and 0.157 nm could be assigned to the (111), (220), and (200) plane of MnO, respectively. These results indicated that small Co/MnO_x nanoparticles were produced during the partial pyrolysis of CoMn-MOF-74. We further compared the electron microscopy of derived samples obtained at 500, 600, 700, and 800 °C to investigate the influence of pyrolyzing temperature on Co/MnO_x nanoparticles. As observed in Fig. S17, with increasing calcination temperature, the size of nanoparticles obviously increased by aggregation. To be noted, the corresponding elemental mapping confirmed the CoMnO_x@C core-shell structure for these completely decomposed samples, as the distribution of O perfectly matched that of Mn on the skin coat of the Co, where C homogeneously distributed in the bulk of high temperature derivatives with the widest profile.

XPS measurements were carried out to illustrate the surface compositions and chemical states of the as-prepared catalysts. The high-resolution Co $2p_{3/2}$ spectra in Fig. 4a reveal the emergence of Co° after heat treatment, interestingly, with the increase of thermal decomposition degree, the surface Co° contents increased from less than 20 % (Co/MnO_x@quasi-MOF-74) to more than 50 % (Co/MnO_x@C-600) (Fig. S18). The Mn 3 s peak was used to help to distinguish Mn oxidation states. As shown in Fig. 4c, for all the samples, the distance between the two peaks of Mn 3 s was above 5.9 eV, which agreed well with that of the MnO [43]. These results further confirmed that Co^{2+} was partially reduced to Co° and Mn element can be remained mainly in bivalent after 4.0 h of pyrolysis, which is consistent with XRD and electron microscopy results. Moreover, the ratio of Co°/(Co° + Co^{2+}) in Co/MnO_x@quasi-MOF-74 sample was estimated to be 18.57 % (Fig. S18), indicating that about 18.57 % of Co species seemed to be decomposed from the framework of CoMn-MOF-74 during heat treatment at 400 °C, along with the formation of Co°. Through comparing the overall (detected by ICP-OES and elemental analyzer, Table S2) and surface (detected by XPS, Table S3) composition of the CoMn-MOF-74 and derived catalysts, it can be seen that the total Co and Mn content in bulk phase increased gradually with the increase of calcination temperature, while the surface content decreased. However, for carbon content, the trend was exactly opposite. These results suggested that carbon was enriched in the surface during pyrolysis, while most of the metals gradually migrated into the interior of the support. Making it eventually forming Co/MnO_x@quasi-MOF-74 core-shell structure after partial pyrolysis, and Co/MnO_x@C core-shell structure at the high temperature due to the complete destruction of the MOF-74 framework.

3.2. Insight into the formation process of Co/MnO_x@quasi-MOF-74 core-shell catalyst

In order to further elucidate the changes in the structure of MOF-74 framework during the heat treatment, we employed TGA (Fig. S19) and temperature-programmed pyrolysis (TPP) measurements (Fig. 20 and S21) to monitor the decomposition profiles of the pristine CoMn-MOF-74. The evolving processes of the CoMn-MOF-74 structure induced by heat treatment were illustrated in Fig. 5. As displayed in Fig. S19, a weight loss about 17.5 wt% between room temperature and 250 °C was attributed to the removal of coordinated solvent molecules (CH_3OH and C_2H_5OH) and water in CoMn-MOF-74 precursor, which created a number of accessible open metal sites (OMSs) or CUSs of Co^{2+} working for catalyzing CO nondissociation and insertion (see below for a detailed explanation). Further weight loss of 15.0 wt% was observed when the temperature was raised from 250 to 400 °C, as evidenced in Fig. S21, a certain amount of reducing gas, such as H_2, $C_{1-4}H_x$, and CO, released from dehydrogenation and further carbonization of the organic ligands, ultimately resulting in a ‘Quasi-MOF-74′. Interestingly, the content of CO in outlet gas was higher than that of CO_2 and increased first but then dropped when the temperature was held at 400 °C for 4.0 h, and we speculate that this might be due to the hydrogenation of CO_2 with reducing gases at metal Co°. In addition, after being kept at 400 °C for 4.0 h, almost no gases were released, indicating that the obtained quasi-MOF structure was in a very stable state, which was also confirmed by TGA results (Fig. S22). A further temperature increased to 600 °C resulted in a dramatic decrease in the weight by 29.8 wt%, implying a complete decomposition of CoMn-MOF-74. What's more, the TGA plots of mono metal (Co, Mn) MOF-74 (Fig. S19) reflected that Mn-O bonding was much stronger than Co-O, which caused that Co was first dissociated with DOT^{4-} ligands and then reacted with reducing gases incorporated in the formwork of MOF-74 to form Co° nanoparticles. However, Mn was likely to be oxidized by the O atoms of DOT^{4-} ligands and decomposed together in the form of MnO_x. These conclusions are well explained how the Co/MnO_x@quasi-MOF-74 structure formed after partial pyrolysis.

3.3. Catalytic results

The catalytic performance of the as-obtained samples for the conversion of syngas were evaluated in a fixed-bed reactor. The reaction

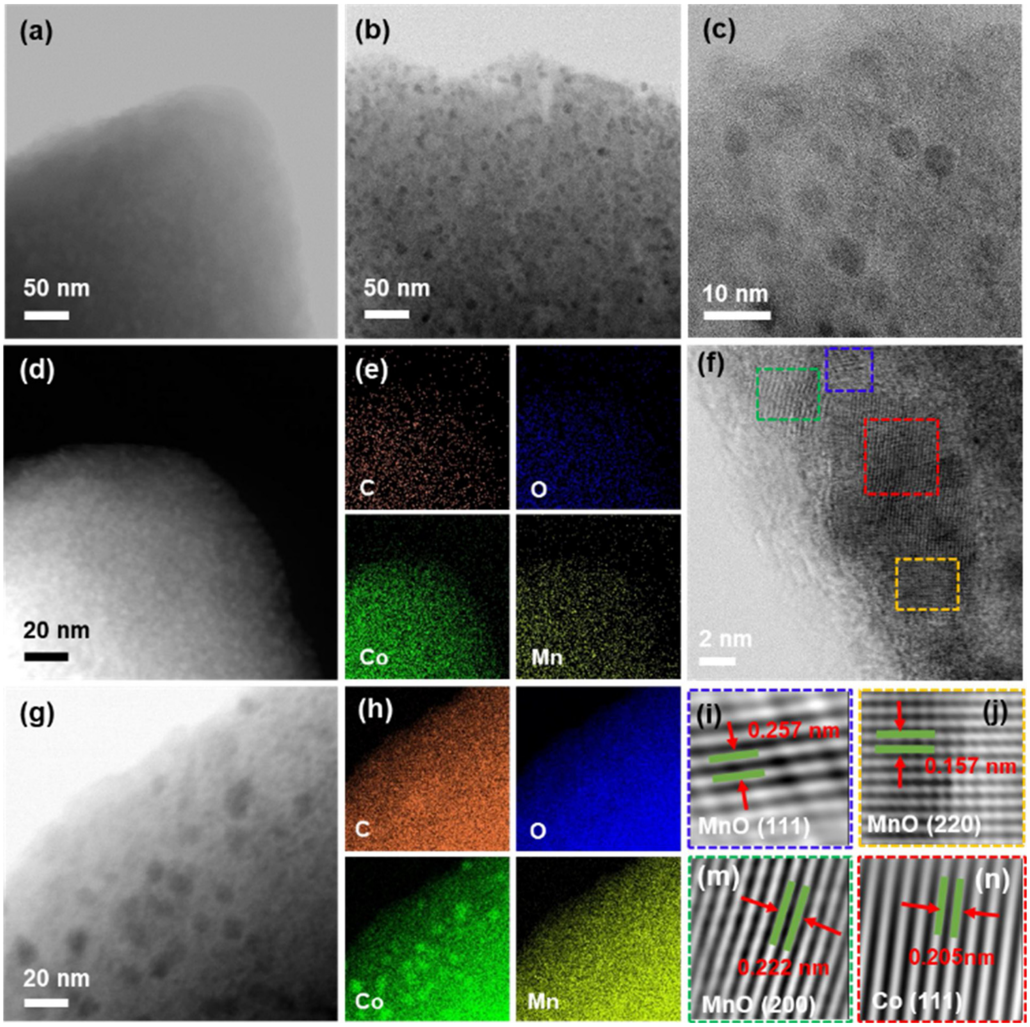

Fig. 3. Electron microscopy characterization. HAABF-STEM images of (a) CoMn-MOF-74 and (b, c) the derived Co/MnO_x@quasi-MOF-74 after controlled partial pyrolysis at 400 °C for 4.0 h. High-resolution TEM images of (d) CoMn-MOF-74 and (g) Co/MnO_x@quasi-MOF-74 and corresponding EDS elemental mappings of (e) CoMn-MOF-74 and (h) Co/MnO_x@quasi-MOF-74, showing the numerous homodispersed nanoparticles emerged after 4 h pyrolysis at 400 °C. (f) High-resolution HAABF-STEM image of Co/MnO_x nanoparticles and the inverse fast Fourier transformation (IFFT) images taken from the selected areas indicated by (i) blue, (j) orange, (m) green, and (n) red, rectangles in (f). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

products with broad carbon distribution consist of gaseous hydrocarbons ($C_{1-4}H_x$, and CO_2, Figs. S4 and S5), ROH (Fig. S6–S8), and liquid hydrocarbons ($C_{5+}H_x$, Figs. S8 and S9). The results of catalytic tests were listed in Table 1, in terms of CO conversion and product selectivity. As expected, the as-obtained Co/MnO_x@quasi-MOF-74 exhibited the best catalytic performance, and the ROH selectivity reached 39.0 wt% (calculations always include CO_2 production) with a CTY of 383.2 $mg_{co}\,g_{cat}^{-1}\,h^{-1}$, where the $C_{2+}OH$ counted for 92.2 wt% of the

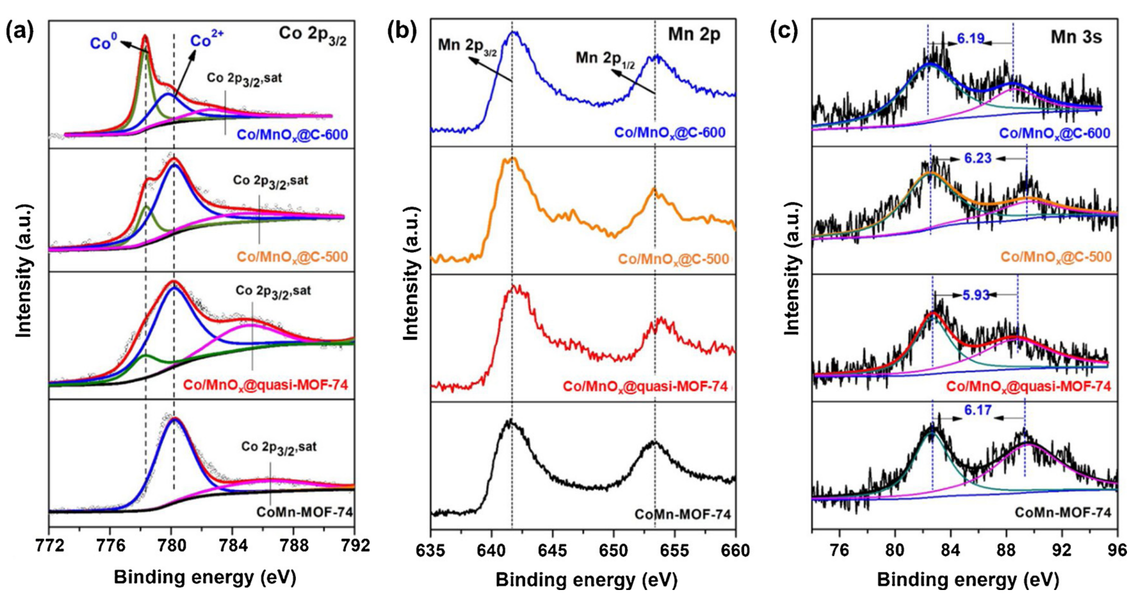

Fig. 4. XPS spectra of Co $2p_{3/2}$ (a), Mn 2p (b) and Mn 3 s (c) of the CoMn-MOF-74, Co/MnO_x@quasi-MOF-74, Co/MnO_x@C-500 and Co/MnO_x@C-600 catalysts.

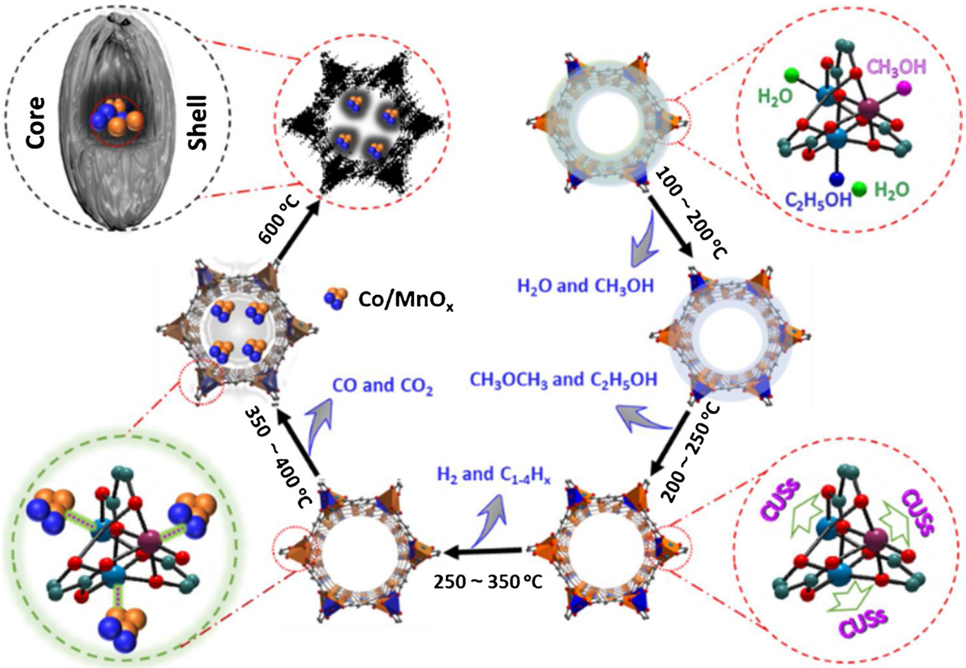

Fig. 5. Schematic illustration of the calcination-induced structure changes of the CoMn-MOF-74. In the initial stage of pyrolysis, namely, at 100-200 °C, the guest H_2O or coordinated H_2O, CH_3OH molecules were released from CoMn-MOF-74. Further increase of calcination temperature to 200-250 °C, a number of CUSs sites were created after complete removal of the coordinated solvent molecules. These accessible CUSs of Co^{2+} sites can nondissociatively adsorbed CO* species and strengthen CO insertion process in HAS. At about 250-350 °C, and 350-400 °C, the partial terephthalate ligand was gradually decomposed and released some reducing gases, including $C_{1-4}H_x$ hydrocarbon, H_2, CO, and CO_2. In this stage, metal Co and Mn were dissociated in the form of Co/MnO$_x$ from the framework of MOF and fabricated a Co/MnO$_x$@quasi-MOF-74 core-shell structure. However, a harsh temperature at 600 °C results in the collapse of the framework. The controlled deligandation of CoMn-MOF-74 produces three types of synergistic active sites (Co°, coordinatively unsaturated sites (CUSs) of Co^{2+} and Co_2C) that collaborate with each other to enhance $C_{2+}OH$ formation during the reaction.

Table 1
Catalytic performances over various catalysts for syngas conversion.

Catalyst[a]	Conv. (%)	Product distribution (wt%)					ROH[e]	$C_{2+}OH$/ROH	ROH + RH=
		CO_2	CH_4	$C_{2+}H$[b]	RH=[c]	$C_{2+}OH$[d]			
Co@quasi-MOF-74	62.0	10.4	8.1	52.4	11.8	14.3	17.3	83.8	29.1
Co/MnO$_x$@quasi-MOF-74	21.4	0.8	10.0	28.6	21.6	35.9	39.0	92.2	60.5
Co/MnO$_x$@C-500	44.5	7.7	21.2	24.9	19.9	23.0	26.3	87.6	42.9
Co/MnO$_x$@C-600	49.9	5.1	29.8	32.6	18.9	10.7	13.4	79.2	29.6
Co/MnO$_x$@C-700	55.9	6.3	46.6	27.7	8.9	7.8	10.5	73.8	16.7

[a] Catalytic tests were conducted at 230 °C, 3.0 MPa, GHSV = 4500 mL g^{-1} h^{-1}, and a H_2/CO ratio of 2. Time on steam (TOS) = 24 h.
[b] $C_{2+}H$: C_{2+} paraffins.
[c] RH=: Olefins.
[d] $C_{2+}OH$: C_{2+} Alcohols.
[e] ROH: Alcohols.

ROH, at 230 °C, 3.0 MPa ($CO/H_2 = 1/2$), and a GHSV of 4500 mL g^{-1} h^{-1}. Meanwhile, among the hydrocarbons (RH) products, the fractions of C_2-C_4 and C_{5+} olefins reached 60.9 and 26.1 wt%, respectively (Fig. 6b), and the total selectivity of high-value added products (ROH and olefins (RH=)) reached 60.53 wt%. More strikingly, it is noteworthy that CO_2 was hardly detectable when using Co/MnO$_x$@quasi-MOF-74, indicating this catalyst has a very low water-gas shift activity. These excellent performances surpass those of state-of-the-art HAS catalysts reported in literatures (Table S7), except for a multi-functional catalyst recently reported by Sun et al., comprised of CoMn oxides and CuZnAlZr oxides [44]. Although this CoMn|CuZnAlZr multifunctional catalyst has a higer ROH selectivity (54.5 wt%), it produces 6.3 wt% of CO_2 and operated at a higher pressure (6.0 MPa) and lower GHSV (2000 mL g^{-1} h^{-1}). In addition, Ma et al. recently report a nano-construct of a Fe_5C_2-Cu interfacial catalyst exhibited a CO conversion of as high as 53.2 %, a selectivity of 49.1 % for long-chain alcohols, but produced more than 30 % of CO_2 [11]. For the completely decomposed catalysts (Co/MnO$_x$@C), they exhibited significantly decreased ROH and RH = selectivity under the same conditions, implied that the well defined pore structure of MOF-74 played a certain role in promoting ROH formation. Control experiments for catalysts without Co (MnO$_x$@quasi-MOF-74) gave negligible alcohol products under the identical conditions, confirming Co as the catalytic center and Mn was the promoter. On the other hand, the pristine CoMn-MOF-74 and Co-MOF-74 showed inferior activities in HAS, which indicated that Co and Mn species in the frameworks were totally inactive for CO activation. The detailed carbon number distributions indicated that the chain length distribution of ROH and hydrocarbons (RH) products followed the Anderson-Schulz-Flory (ASF) statistics, on the other hand, suggesting that the chain growth steps are very similar to those of the well-known Co° catalyzed FTS reaction. However, the chain growth probability (α) values for ROH (α_{ROH}) were below to that for RH (α_{RH}) at all operating conditions in this study, as shown in Fig. 6c and d and Tables S4 and S5, suggested that formation of these two types of products may either proceed on different active sites or by different mechanisms [15].

Very interestingly, the CO conversion and paraffin selectivity, especially for C1 products, increased with increasing calcination temperature of CoMn-MOF-74 precursor, while the total selectivity of ROH and olefins showed the opposite trend (Table 1 and Fig. 6a, b). For example, Co/MnO$_x$@C-700 showed 55.9 % CO conversion and the RH selectivity reached above 50 wt% (36.8 wt% is accounted for CH_4) with only 10.5 wt% of ROH and olefins was obtained at 230 °C, 3.0 MPa, CO/H_2 = 1/2, GHSV = 4500 mL g^{-1} h^{-1}. Knowing that Co° is usually responsible for H_2 and CO dissociative adsorption and carbon chain growth in HAS or FTS, in other words, a higher Co° concentration would give a higher CO conversion under certain conditions [45]. As evidenced by XPS characterization mentioned above, the content of Co° species increased with increasing calcination temperature (Fig. 4 and

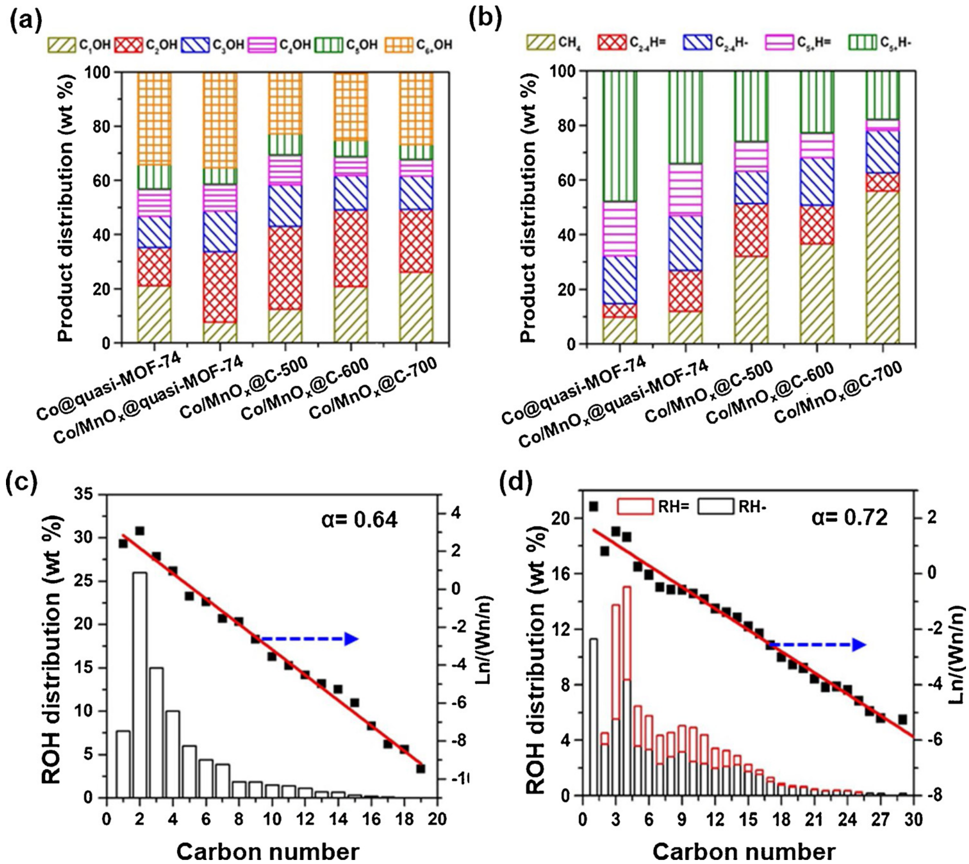

Fig. 6. The ROH (a) and hydrocarbons (b) distribution of various MOF-74 derived catalysts, and ASF plots for ROH (c) and hydrocarbons (d) over Co/MnO$_x$@quasi-MOF-74. Reaction conditions: 230 °C, 3.0 MPa, CO/H$_2$ = 1/2, GHSV = 4500 mL·g^{-1} h^{-1}.TOS = 24 h.

Fig. S18), which could give a good explanation for this tendency. In the case of Co@quasi-MOF-74, up to 61.98 % CO conversion and the highest α_{RH} value (0.73, Fig. S24) were achieved at the identical conditions, which may further conform this conclusion. However, the opposite tendency of ROH also implied the CO nondissociatively adsorbed sites for CO insertion within these catalysts decreased with the increasing of calcination temperature, which will be discussed in the following paragraphs.

In addition, we examined the effects of GHSV and temperature on the catalytic performance of Co/MnO$_x$@quasi-MOF-74 catalyst. As shown in Table S4, increasing the GHSV from 4500 to 15,000 mL g^{-1} h^{-1} reduced the CO conversion to 10.49 % but increased the CTY of CO and space time yield (STY) of high-value added products (C_{2+}OH and RH=) to 619.9 mg$_{CO}$ g$_{cat}$$^{-1}$ h^{-1} and 206.0 mg$_{ROH+RH}$ g$_{cat}$$^{-1}$ h^{-1}, respectively. Meanwhile, the C1 products had hardly changed and the total selectivity of ROH and RH = kept in the range of 57 − 61 wt% (Table S4). These results essentially demonstrated that high GHSV, on the one hand, to a certain extent, promoted the conversion rate of CO and the ROH formation rate, on the other hand, restrained the formation of CH_4, CO_2 and CH_3OH. This phenomenon is contrary to literature studies [15]. It is generally recognized that the molar selectivities to low carbon products, especially for CH_4 and CH_3OH, would be expected to be relatively high at a high GHSV, just as the results of using Co/MnO$_x$@C-600 catalyst (Table S6), since these products were formed initially in the reaction network and low GHSV will give enough time for the C—C coupling or subsequent CO insertion [15,45]. Definitely, both the values of α_{ROH} and $\alpha_{RH,}$ however, showed a decrease with the increasing of GHSV (Table S4), which was consistent with the same reaction network described above. We therefore speculated that the porous formwork of quasi-MOF-74 can increase inward diffusion rate of some lower carbon molecules but limit the long-chain molecules enter and outer of the pore channels thereby leave more active sites for facilitating C1 molecules transformations. Meanwhile, it can be also seen from Table S5 that ROH formed with highest selectivity (48.7 wt%, and 93.1 wt% was C_{2+}OH) and almost no gaseous CO_2 was detected under conditions of low CO conversion (6.7 %, a CTY of 117.8 mg$_{CO}$ g$_{cat}$$^{-1}$ h^{-1}) at 200 °C.

The stability of the Co/MnO$_x$@quasi-MOF-74 catalyst was tested by running the reaction for 120.0 h (at 230 °C for the first 50.0 h and 250 °C for the last 70.0 h). As shown in Fig. 7, the CO conversion decreased rapidly within the first 10.0 h, and, afterward, basically remained unchanged at about 12 % (230 °C) and 30 % (250 °C) during the reaction. No sign of deactivation was observed at 120.0 h, and the selectivity toward gaseous (C_1-C_4, CO_2) and liquid (C_{5+}) products also did not undergo dramatical changes throughout the test, indicating that Co/MnO$_x$@quasi-MOF-74 catalyst has a superior stability.

3.4. Investigation of the active sites

Co° is a well-known active site for hydrocarbon formation in HAS due to its strong CO dissociation ability [1,5–7,18,22,45]. However, to get a higher C_{2+}OH selectivity, other synergetic sites are generally needed to facilitate the nondissociative CO insertion for the formation of C_{2+}OH [1,3,9,12,13,22]. Inspired by this, we first performed in situ diffuse reflectance infrared Fourier transform spectroscopy (CO-DRIFTS) experiments to investigate the nature of the active sites of Co/MnO$_x$@quasi-MOF-74 catalyst. As shown in Fig. 8a, the spectrum of the pristine CoMn-MOF-74 was dominated by a relatively intense peak located at 2163 cm^{-1}. More importantly, this peak possessed a narrower band and remained observable even when the temperature up to 250 °C, suggesting an electronically uniform and stable set of Co − CO coordination sites [46]. The peaks at 2120 − 2170 cm^{-1} could be

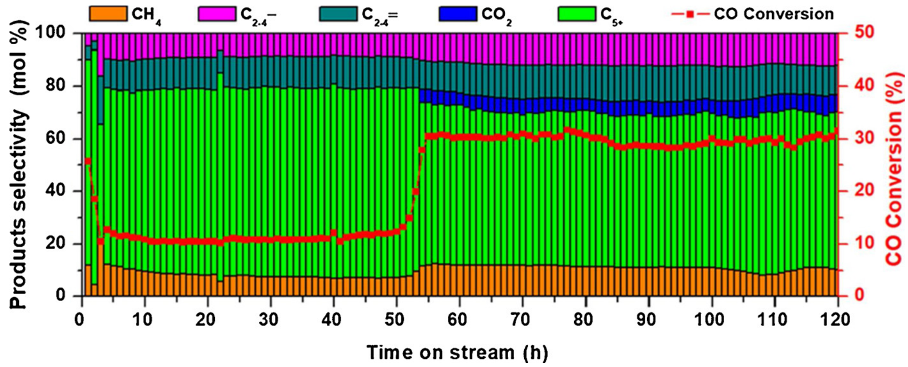

Fig. 7. CO conversion and product selectivities as a function of time on stream at 230 °C (0-50 h) and 250 °C (51-120 h). Other condition: P = 3.0 MPa, CO/H_2 = 1/2, GHSV = 15,000 mL·g·h^{-1}.

assigned to linearly adsorbed CO on Co^{2+} [5,47–49]. Thus, it is reasonable to conclude that the peak around of 2163 cm^{-1} can be assigned to the CO adsorption on CUSs of Co^{2+} in the formwork of MOF-74. Interestingly, the intensity of the 2163 cm^{-1} peak related to CO-Co^{2+} sites on Co/MnO_x@quasi-MOF-74 catalyst was much lower than that on pristine CoMn-MOF-74 and disappeared in the completely decomposed samples (Co/MnO_x@C-500, 600). This may be because that the portion Co of the MOF-74 skeleton was detached into the interior of the skeleton in the process of partial pyrolysis, and lowered the density of the skeleton Co, thereby causing a significant decrease in the Co^{2+}-CO intensity. An obvious red shift of the peak at 2159 cm^{-1} can be observed for Co/MnO_x@quasi-MOF-74 in reference to that of CoMn-MOF-74, which might be due to that, as the concentration of CUSs of Co^{2+} decreases after partial pyrolysis, the surface electron density for the CO-coordinated Co^{2+} sites increased [46,50]. However, for the completely decomposed samples, the Co^{2+}-CO peak disappeared completely due to the complete collapse of the skeleton. In previous studies, Co°-Co^{2+} pairs, or to say, Co^{2+} sites, have proven to be effective for non-dissociative CO adsorption and also can enable significant CO insertion to occur in HAS [5,22]. Therefore, the high density and evenly dispersed Co^{2+}-CO active sites on the formwork of MOF-74 can enable CO insertion into a metal-alkyl bond, which may be accounted for the exceptional behavior of Co/MnO_x@quasi-MOF-74 catalyst. In addition, there were two strong absorbance peaks located at 2029 and 1967 cm^{-1} that were also observed at 30 °C on all CoMn-MOF-74 derived catalysts, which could be assigned to the species of CO linearly adsorbed and bridge-adsorbed on Co° sites. Obviously, compared with Co^{2+}-CO species, these Co°-CO species were more unstable, which could be activated at a high temperature with the breaking of C—O bond.

In addition, it is known that Co^0 is easily transformed to Co_2C phase under realistic FTS, and the resulted Co_2C is also efficient for CO*/CH_xO* insertion to produce C_{2+} alcohols [6–8,44,51]. Thus, we further investigated the structural information of spent Co/MnO_x@quasi-MOF-74 catalyst to observe whether a new Co_2C phase was formed. Combined XRD patterns and HAABF-STEM images (Fig. 8b, c), a mixture of Co^0 and Co_2C phases was identified in the spent catalyst. We infer that the Co_2C nanoparticles formed during HAS may provide another active site, similar to that of the CUSs of Co^{2+}, which was responsible for CO nondissociative adsorption and CO insertion as it also was evidenced by

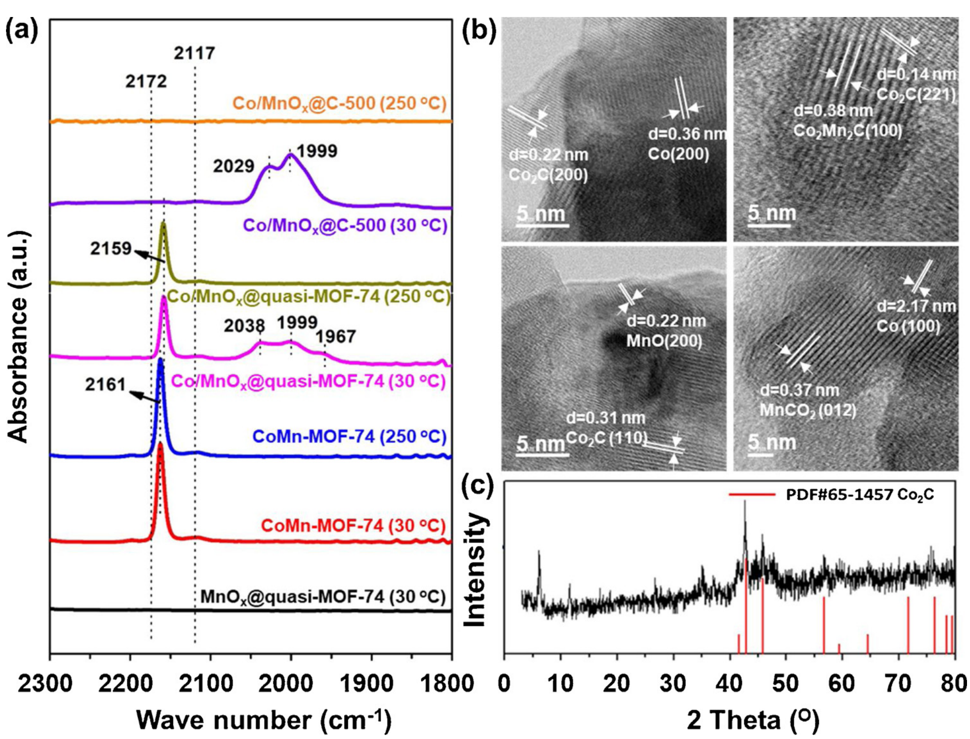

Fig. 8. (a) DRIFTS spectra of CO adsorption on MnO_x@quasi-MOF-74, CoMn-MOF-74, Co/MnO_x@quasi-MOF-74, and Co/MnO_x@C-500 at 30 °C and 250 °C, respectively. (b) HAABF-STEM images and (c) XRD of spent Co/MnO_x@quasi-MOF-74.

a certain amount of ROH yield in Co/MnO_x@C catalysts. According to these experimental observations, we can conclude that, after partial deligandation, the as-obtained Co/MnO_x@quasi-MOF-74 catalyst generated a large amount of accessible Co^0-CoC_2-Co^{2+} synergistic active centers for the HAS from syngas. On the one hand, Co^0 can effectively catalyze the dissociation of CO and promote the formation of CH_y* intermediate, followed by carbon chain growth to form C_xH_y* species, as indicated by the dramatic increase in the CO conversion and RH selectivity with the increase of calcination temperature (which resulted in an increase of the Co^0 content). On the other hand, the CUSs of Co^{2+} and Co_2C dual sites were responsible for nondissociative CO adsorption and subsequent CO insertion into the C_xH_y* species to form ROH. To the best of our knowledge, never had been reported in literature for such dual CO insertion sites catalysts in HAS. In other words, the CUSs of Co^{2+} in the formwork of quasi-MOF-74, compared to Co/MnO_x@C catalyst, further boosted the CO* (or CH_xO*) insertion in this catalytic system, which is also the main reason why Co/MnO_x@quasi-MOF-74 catalyst has such an outstanding $C_{2+}OH$ selectivity.

3.5. The role of Mn in the Co/MnO_x@quasi-MOF-74 catalyst

Based on the catalytic results, it can be seen that the presence of Mn played an important role during the HAS process. The Co@quasi-MOF-74 catalyst showed significantly higher CO conversion and alkanes selectivity than the Co/MnO_x@quasi-MOF-74. In order to distinguish the contribution of Mn in the Co/MnO_x@quasi-MOF-74, we used XPS and in situ CO-DRIFTS measurements to investigate how the Mn affected the chemical environment of the Co species. As shown in Fig. 9a, compare with Co/MnO_x@quasi-MOF-74, the Co2$p_{3/2}$ peak position (778.75 eV) attributable to Co^0 species in the catalyst without Mn (Co@quasi-MOF-74) shifted to a lower binding energy (778.25 eV), indicating that there are strong interaction between Co and MnO and they do not exist in Co@quasi-MOF-74 [42,52]. As expected, the ratio of $Co^0/(Co^0+Co^{2+})$ in Co/MnO_x@quasi-MOF-74 sample was much lower than Co@quasi-MOF-74 (Fig. 9a and S18), which probably corresponds to the higher CO conversion [22]. Meanwhile, as shown in Fig. 8b, the peak of CO-Co^{2+} (2162 cm^{-1}) shifted to lower wavenumber (2159 cm^{-1}). The red shift of the IR bands indicates an increased electron donation to the CO π* orbital [53]. The catalyst in presence of Mn exposed lower surface Co atoms, leading to a decreased localization of the valence electrons. This was an evidence that the introduction of Mn may dilute or extend the spatial distance of adjacent Co^{2+} sites in the frameworks of MOF-74, which can prevent metal agglomeration and favorably construct isolated Co^{2+} sites during the pyrolysis of MOFs [54]. On the other hand, Mn is often used as a promoter in Co based FTS catalyst to promote the formation of Co_2C during the FTS or HAS process, facilitating the CO insertion [8,44,51,55]. These results confirmed the efficient Mn modification to Co on the Co/MnO_x@quasi-MOF-74 catalyst, and were possibly the reasons why the addition of Mn dramatically enhanced the selectivity of alcohols while inhibit alkanes formation.

3.6. Possible reaction pathway of CO hydrogenation to $C_{2+}OH$

We characterized Co/MnO_x@quasi-MOF-74 catalyst pretreated with 10 % H_2-90 % N_2 at 300 °C for 2.0 h H_2 by the semi-in situ XPS method. The results revealed that, after H_2 pretreatment, there were no obvious changes observed in the resulting XPS spectra (Figs. S25 and S26), which suggested that both the Co and Mn species on the surface of Co/MnO_x@quasi-MOF-74 can not be reduced. The experimental results of the H_2-TPR also supported this reasoning, as shown in Fig. S27, no reduction peak of the metal oxide was identified in the temperature range of 50 − 300 °C. Meanwhile, this result is just more evidence for the high stability of the CUSs of Co^{2+} in the formwork of quasi-MOF-74. Then, we treated Co/MnO_x@quasi-MOF-74 with a mixed gas H_2/CO = 2, at 230 °C, 0.1 MPa, without exposure to air after the H_2 pretreatment. The result revealed that the Co species still coexisted in the form of Co°/Co^{2+} at reaction condition, which in turn confirmed the suggestion of CO-DRIFTS. Furthermore, the content of surface Co^{2+} species (6.79 atom%) slightly decreased after reacting with the reaction gas (to 3.34 atom%, Fig. S26), while the CH_2OH* groups of C species increased from 22.69 to 25.89 atom% (Fig. 10a and S28), suggesting some of CO molecules indeed adsorbed on the CUSs of Co^{2+} and were converted into CH_xO* or OH* species on the catalyst surface (see below for a detailed explanation). As shown in Fig. 10b and c, the TPD results demonstrated that the amounts of CO and H_2-adsorbed on these catalysts were in a decrease order of CoMn-MOF-74 > Co/MnO_x@quasi-MOF-74 > Co/MnO_x@C-500 > Co/MnO_x@C-600. Especially on the Co/MnO_x@quasi-MOF-74 catalyst, the adsorbing capacity of CO much higher than those of Co/MnO_x@C catalysts. This is chiefly because that the formwork of MOF-74 processes exceptionally high specific surface area, pore structure and concentration of coordinatively unsaturated metal surface sites, which can interact strongly with small gas molecules such as H_2 and CO [56,57], leading a higher uptake on CoMn-MOF-74 and Co/MnO_x@quasi-MOF-74 than those of Co/MnO_x@C catalysts. The above results verified the suggestions of CO-DRIFTS experiments.

To further gain detailed information on the possible intermediates during this HAS process, we performed an in situ Raman spectroscopy (Fig. S29) and FT-IR (Fig. 11 and S30) experiments on the pretreated Co/MnO_x@quasi-MOF-74 catalyst at 230 °C, 0.1 MPa, 30 mL·min^{-1} of H_2/CO = 2. A number of reaction intermediates were observed and

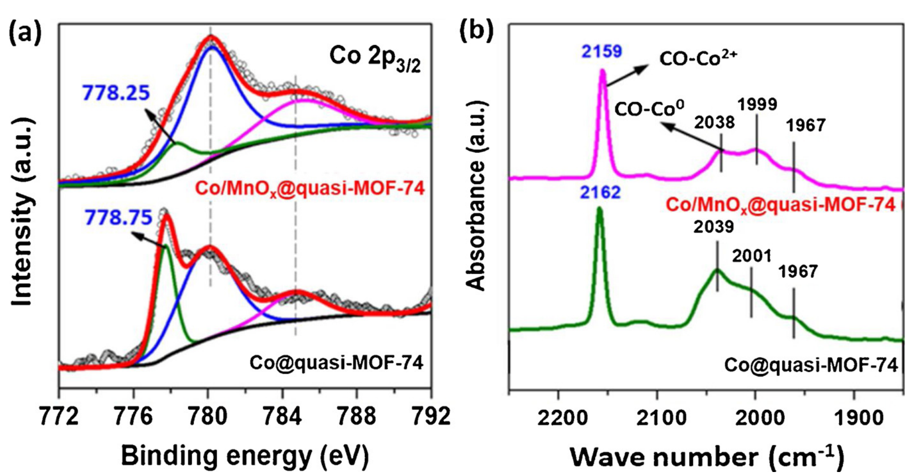

Fig. 9. (a) XPS spectra of the catalysts at Co $2p_{3/2}$ core levels; (b) In situ DRIFTS spectra of CO adsorption on the reduced catalysts.

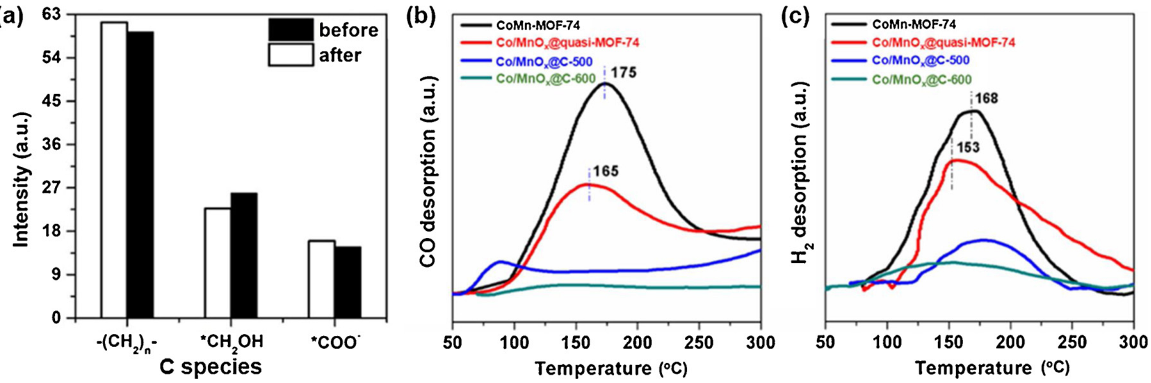

Fig. 10. (a) The content of various C species of the Co/MnO_x@quasi-MOF-74 catalyst observed from semi-in situ C1 s XPS spectrum before and after 2.0 h of reaction. (b) CO-TPD and (c) H_2-TPD data of the catalysts. The TPD signals had been normalized to the mass of the tested samples.

identified, mainly including HCOO*, H_3CO*, OH* and CH_2/CH_3* species. In the initial stage of the reaction, the Co/MnO_x@quasi-MOF-74 catalyst exhibited very high activity and the CO was almost completely converted. At the same time, three type of strong peaks at 2900–3000, 1620 − 1560 and 1450 − 1470 cm^{-1} were observed. The peak at 2967 cm^{-1} was attributed to a combination of the ν_{as}(OCO) and δ(CH) modes for the formate species [58], while the feature peaks at 1586 and 1362 cm^{-1} were assigned to the antisymmetric ν_{as}(OCO) and ν_s(OCO) of adsorbed bidentate HCOO* species [43,59]. The bands at 1468 and 1456 cm^{-1} were assigned to the C—H bending δ(C—H) and stretching vibrations ν_s(C—H) of the CH_2* species [43,49,59,60]. Moreover, the formation of H_2O was also confirmed by the increase of the peak at 1633 and 1645 cm^{-1}. Notably, after 4 min of reaction, the gaseous CO_2 or adsorbed CO_2 (CO_{ad}) (2343 and 2353 cm^{-1}) was vanished. The results indicated that CO was firstly hydrogenation into HCOO* and CH_2/CH_3* species under H_2 atmosphere. Furthermore, it can be seen that the HCOO* species were gradually reduced to hydroxyl groups (OH*, 3675, 3589 and 1633 cm^{-1}) and/or methoxy group (CH_xO*, 1683 and 1613 cm^{-1}) by H atoms with the increase of reaction time. With the consumption of HCOO* and CH_2/CH_3*, the peaks of OH* and CH_xO* became more and more remarkable, which further proved such conversion. Two broad peaks at 3675 and 3598 cm^{-1} correspond to two distinct OH-stretch vibrations, indicating that the OH* species experience distinct environments at the surface or within the pores of Co/MnO_x@quasi-MOF-74 catalyst. The peak at higher wavenumber is associated with the presence of free OH bonds, i.e., OH bonds that are not engaged in H-bonding, while the peak at lower wavenumber (3598 cm^{-1}) with a narrower band can be attributed to the stretching vibration of the OH bond weakly interacting with the CUSs of Co^{2+} (Co^{2+}—OH) [61]. However, as shown in Fig. S30, it was clear that, for the totally decomposed MOF-74 derivative (Co/MnO_x@C-600), there is only one broad peak at around 3500 cm^{-1}, which was a further

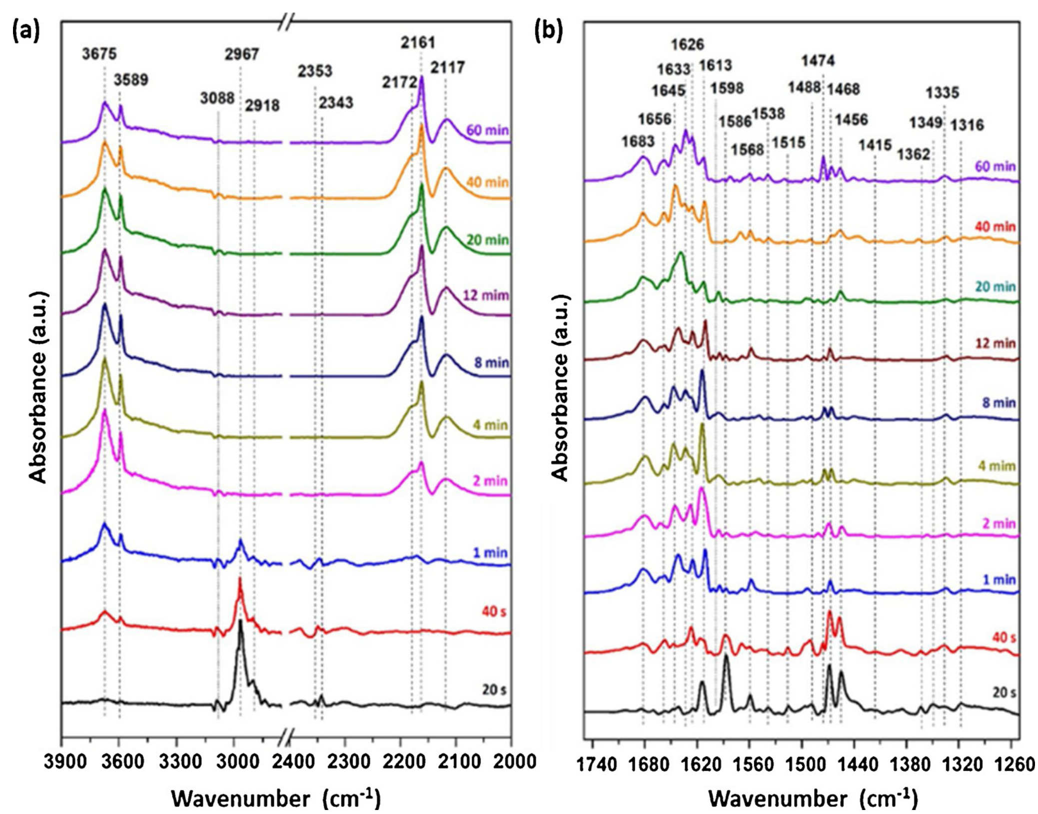

Fig. 11. In situ DRIFT spectra of the CO hydrogenation reaction on the pretreated Co/MnO_x@quasi-MOF-74 catalyst in the regions (a) 3900-2000 cm^{-1} and (b) 1740-1260 cm^{-1} (Reaction conditions: 10 mL·min^{-1} CO + 20 mL·min^{-1} H_2, 0.1 MPa, 230 °C).

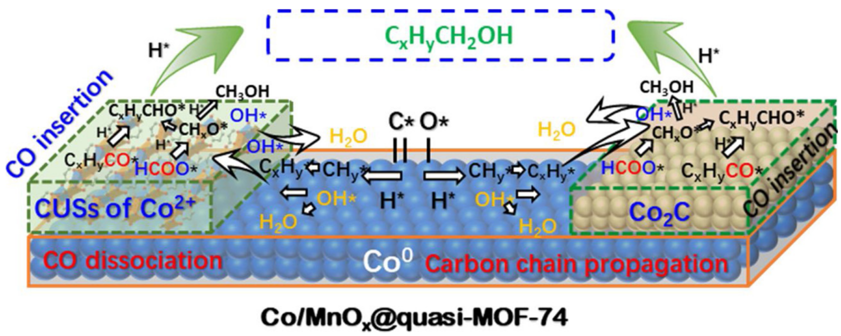

Fig. 12. The active sites and possible reaction pathway. Depending on the structural characteristics of MOF-74, the obtained Co/MnO$_x$@quasi-MOF-74 catalyst possessed three types of synergistic active sites, namely Co^0, CUSs of Co^{2+} and Co_2C, that collaborate with each other to enhance $C_{2+}OH$ formation in HAS process. The Co^0 nanoparticles within the framework of quasi-MOF-74 enable CO dissociation and significant CH_x-CH_y coupling to occur while the uniformly distributed CUSs of Co^{2+} working with Co_2C strengthen CO insertion process, leading an outstanding catalytic performance in the process of CO hydrogenation. The products distribution followed the ASF law, and the HAS process followed CO insertion mechanism.

indication of the existence of Co^{2+}-OH species on the CUSs of Co^{2+} for Co/MnO$_x$@quasi-MOF-74 catalyst during the reaction.

According to the catalytic performance and characterization results above, we propose a possible reaction mechanism as shown in Fig. 12. Firstly, a part of CO and H_2 adsorbed and dissociated into C*, O* and H* species on the surface of Co^0 nanoparticles and hydrogenated to CH_y* and OH* species or H_2O, followed by carbon chain growth to form C_xH_y*, while the other part of CO nondissociatively adsorbed on the CUSs of Co^{2+} formed CO* intermediates. The product distribution followed the ASF statistics, indicating that the chain growth steps were very similar to those of the well-known Co^0 catalyzed FTS reaction. Secondly, the produced C_xH_y* and OH* species migrated to the CUSs of Co^{2+} or Co_2C active sites, where the nondissociatively adsorbed CO* species were inserted into these alkyl chain to generate $C_{2+}OH$. On the other hand, OH* reacted with other CO* to form HCOO* intermediates, subsequently stepwise hydrogenated to CH_xO* and OH* intermediates, the transformation of OH* intermediates, therefore, forming a circulation (for details see Fig. S31). In addition, CH_xO* species (mainly consist of CHO*, CH_2O* and CH_3O*) were generally recognized as other intermediates that could more preferentially inserted into C_xH_y* than CO*, which would also undergo insertion reactions with alkyl species to form C_xH_yCHO* species and further was reduced to $C_{2+}OH$ [14,44,45]. In a word, the synergetic catalysis of confined Co^0 nanoparticles and uniformly distributed CUSs of Co^{2+} and Co_2C active sites were responsible for synthesis of ROH over the Co/MnO$_x$@quasi-MOF-74 catalyst.

4. Conclusions

In summary, it is demonstrated that the controlled deligandation of a robust bimetallic CoMn-MOF-74 by partially thermal treatment strategy can fabricate a core-shell Co/MnO$_x$@quasi-MOF-74 catalyst. This catalyst not only largely maintained the framework of MOF-74 for effective substrates diffusion, but also in situ generated a number of highly active Co/MnO$_x$ nanoparticles inside quasi-MOF-74, made it provide three types of synergistic active sites (Co°, CUSs of Co^{2+} and Co_2C) in the process of HAS. Benefiting from the appropriate cooperation between the created Co° nanoparticles, which are responsible for CO dissociation and subsequent carbon chain growth, and highly dispersed, adjacent CUSs of Co^{2+} and Co_2C, which are active for CO non-dissociative adsorption and insertion, Co/MnO$_x$@quasi-MOF-74 catalyst exhibited a dramatically enhanced catalytic performance in the HAS. The selectivity to alcohols (ROH) reached as high as 48.7 wt% with a CTY of 117.8 $mg_{co}\,g_{cat}^{-1}\,h^{-1}$, where 93.1 wt% was $C_{2+}OH$, at 200 °C, 3.0 MPa ($CO/H_2 = 1/2$), and a GHSV of 4500 mL $g^{-1}\,h^{-1}$, which outperforms those of completely decomposed Co/MnO$_x$@C, and state-of-the-art CO hydrogenation catalysts reported in literatures. More significantly, very low CH_4 and indetectable CO_2 were produced during the reaction. This work highlights opportunities in using MOF-derived quasi-MOFs materials as novel supports for designing highly effective catalysts with multi active sites.

CRediT authorship contribution statement

Wen-Gang Cui: Conceptualization, Methodology, Visualization, Investigation, Writing - original draft. **Yan-Ting Li:** . **Hongbo Zhang:** Writing - review & editing. **Zheng-Chang Wei:** Investigation. **Bo-Hai Gao:** Investigation. **Jing-Jing Dai:** Investigation, Visualization. **Tong-Liang Hu:** Supervision, Writing - review & editing, Conceptualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare no competing financial interests.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.apcatb.2020.119262.

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Supporting Information

In situ encapsulated Co/MnO_x nanoparticles inside quasi-MOF-74 for the higher alcohols synthesis from syngas

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Table of Contents

S1. Experimental section

S1.1. Materials

All the reagents and solvents used in this study were analytical grade and used directly without any further purification. Cobalt nitrate hexahydrate ($Co(NO_3)_2·6H_2O$) and 2,5-dihydroxyterephthalic acid (H_4DOT) were obtained from Meryer (Shanghai) Chemical Technology Co., Ltd. Manganese chloride ($MnCl_2·4H_2O$) was obtained from Aladdin Industrial Corporation, China. Ethanol, methanol and *N,N*-dimethylformamide were obtained from Tianjin Hengshan Chemical Technology Co. The syngas $H_2/CO/N_2$ = 63.5/31.5/5.0, N_2 (99.99%), Ar (99.999%) and H_2 (99.99%) were provide by Air Liquide (Tianjin) Co., Ltd.

S1.2. Catalyst Preparation

Synthesis of Co-MOF-74. $Co(NO_3)_2·6H_2O$ (0.713 g, 2.450 mmol) and H_4DOT (0.20 g, 1.009 mmol) were dispersed in 60 mL of a mixed solvent (DMF/ethanol/water = 1:1:1 (v/v/v)), put into a 100 mL Teflon-lined autoclave, stirred well and kept at 120 °C for 24.0 h. Then it was cooled to room temperature, and the powder product was washed with fresh MeOH for 6-8 times over a 3 days period, dried at 40 °C under vacuum overnight.

Synthesis of Mn-MOF-74. $MnCl_2·4H_2O$ (0.66 g, 3.335 mmol) and H_4DOT (0.20 g, 1.009 mmol) were dissolved in a mixed solvent (DMF/ethanol/water = 55.0/3.5/3.5 mL), put into a 100 mL Teflon-lined autoclave, kept at 135 °C for 24.0 h, and then cooled to room temperature. The resulting powder was washed with fresh MeOH for 6-8 times over a 3 days period, dried at 40 °C under vacuum overnight.

S2. Supplementary Figures

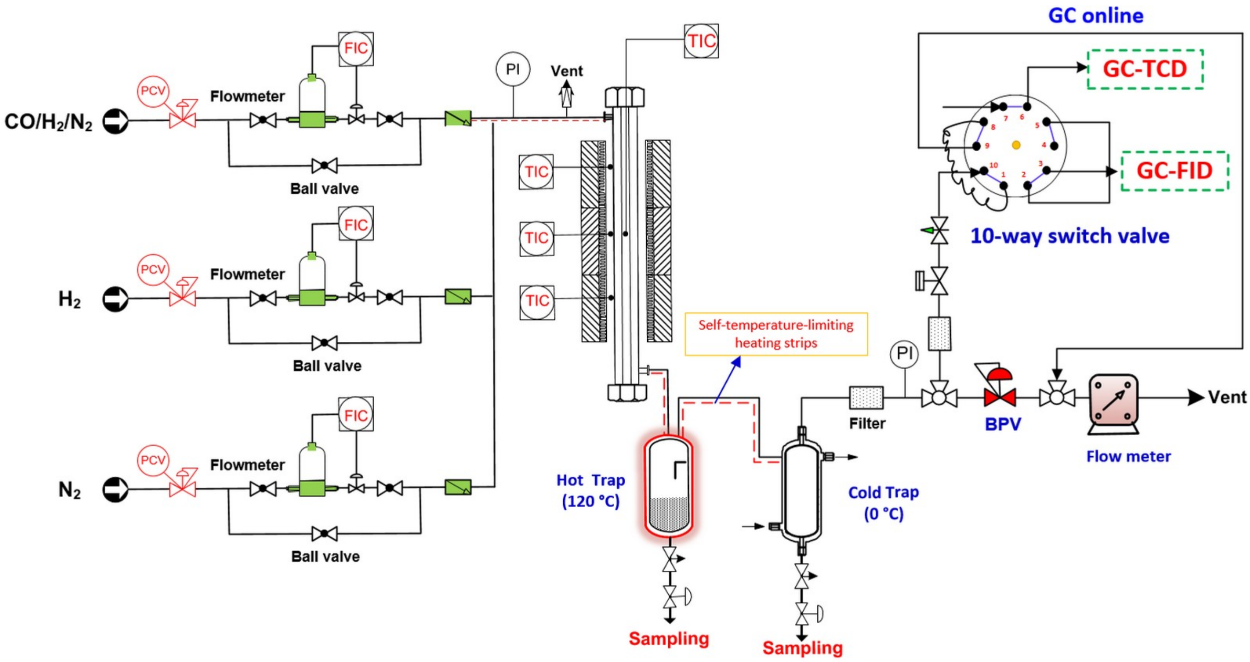


Figure S1. Schematic diagram of the reaction system for syngas conversion. PI: pressure indicator; TIC: temperature indicator control; BPV: back pressure valve; BCV: pressure control valve; FIC: flow indicating controller.

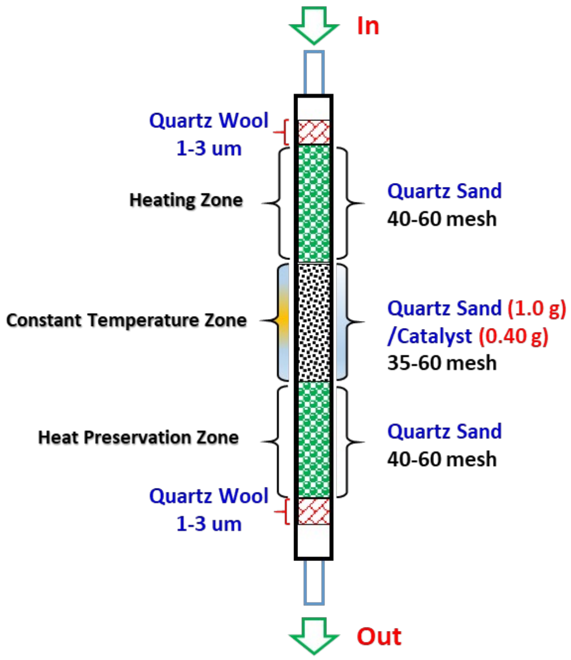


Figure S2. Illustration of loading of catalyst in reactor.

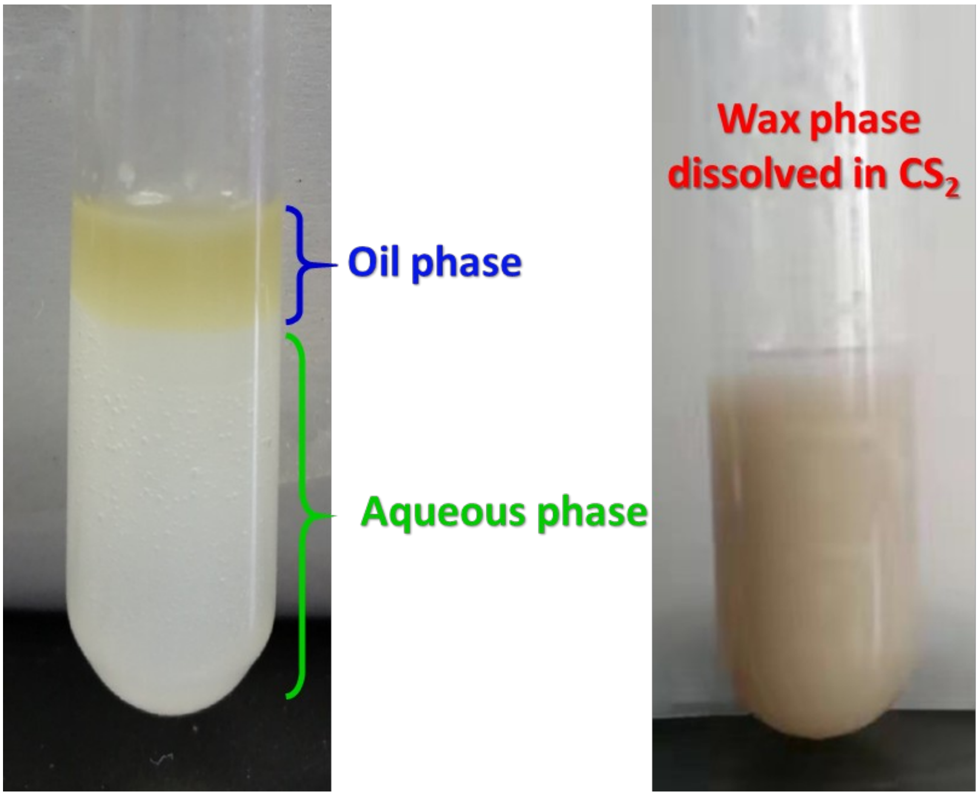


Figure S3. Optical photographs of liquid product obtained from cold trap and hot trap. It consists mainly of three phases (oil phase, aqueous phase (left) and wax phase (right, dissolved in carbon disulfide)).

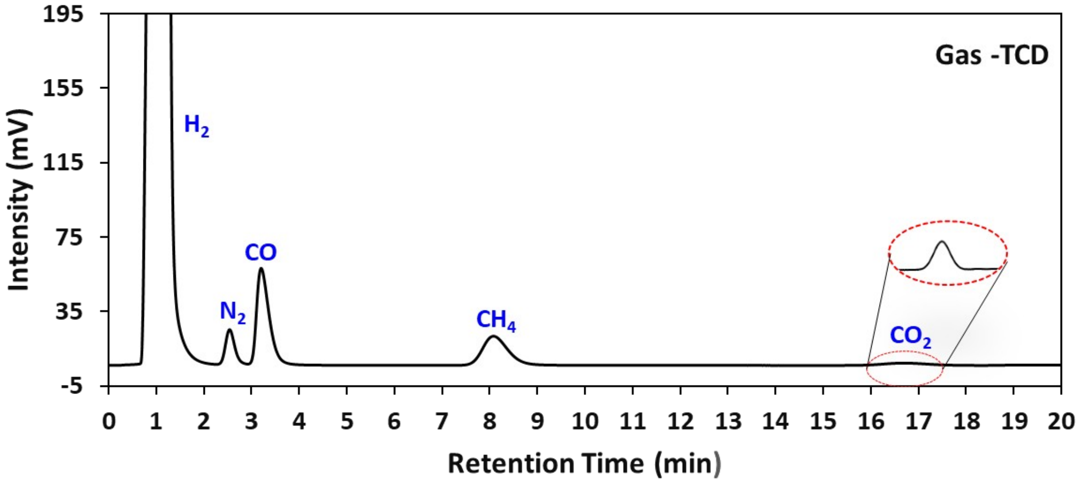


Figure S4. Typical GC-TCD spectrum of gas products.

Gas Chromatographic Conditions: Chromatographic column: TDX-01 packed column; Injection temperature: 120 °C; Detector temperature: 150 °C; Column temperature: 100 °C kept for 19.0 min; Carrier gas: Ar (purity > 99.999%). Quantitative method: External standard method using the peak areas. Instrument model: Huifen GC-6890; Sampling method: On-line.

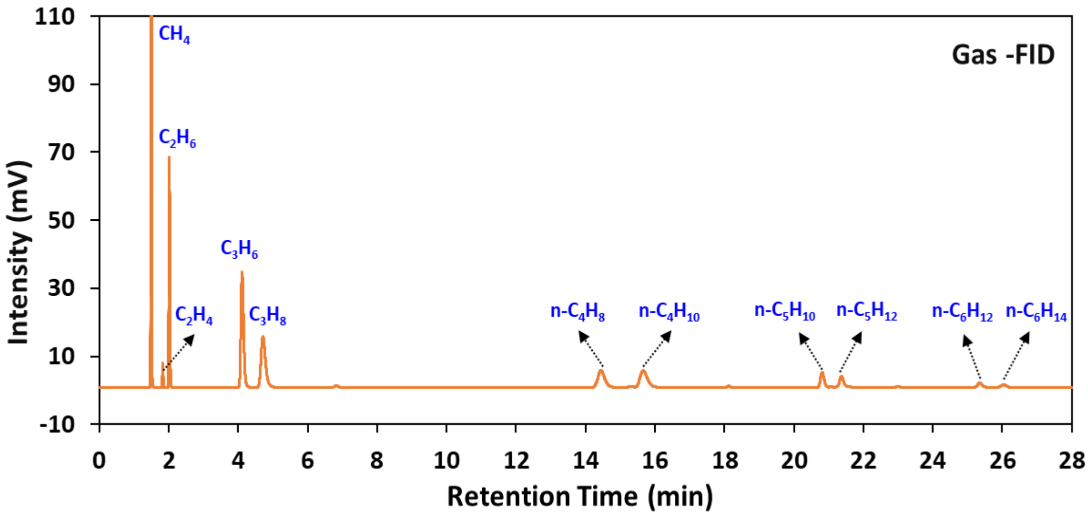
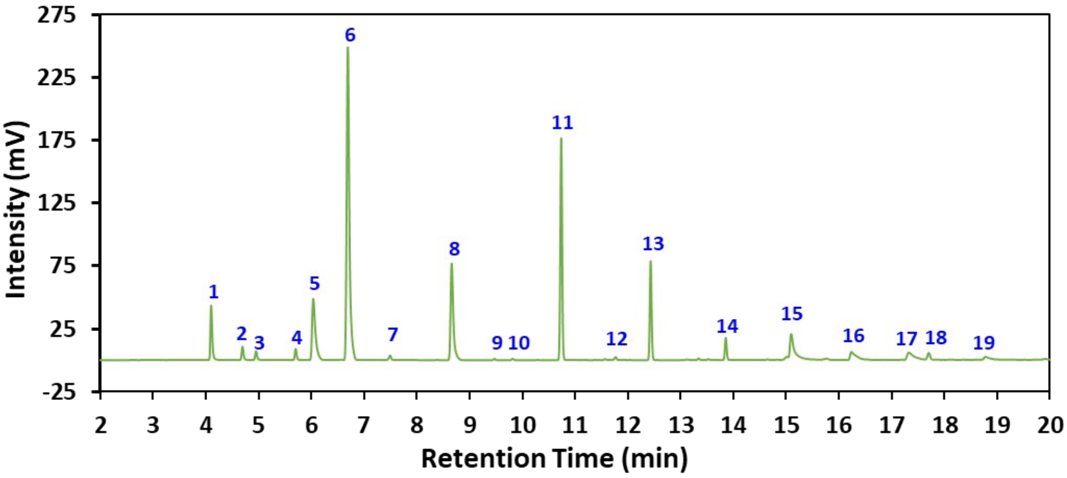


Figure S5. Typical GC-FID spectrum of gas products.

Gas Chromatographic Conditions: Chromatographic column: HT-PLOT Q (30 m ⁎ 0.53 mm ⁎ 40 um) capillary column; Injection temperature: 120 °C; Detector temperature: 220 °C; Column temperature: 100 °C kept for 19 min, then rise up to 200 °C at a speed of 10 °C/min; Carrier gas: Ar (purity > 99.999%). Quantitative method: External standard method using the peak areas. Instrument model: Huifen GC-6890; Sampling method: On-line.



Peak	Compound	Peak	Compound	Peak	Compound	Peak	Compound
1	Ethanal	**6**	Ethanol	**11**	n-Butanol	**16**	Acetic acid
2	Propanal	**7**	Pentanal	**12**	Isoamylol	**17**	Propionic acid
3	Acetone	**8**	Propanol	**13**	Pentanol	**18**	Butyric acid
4	n-Butanal	**9**	Hexanal	**14**	*sec*-Pentanol	**19**	Pentanoic acid
5	Methanol	**10**	*sec*-Butanol	**15**	n-Hexanol		

Figure S6. Typical GC-FID spectrum of aqueous phase products.

Gas Chromatographic Conditions: Chromatographic column: DB-WAX (30 m * 0.25 mm * 0.25 um) capillary column; Injection temperature: 250 °C; Detector temperature: 260 °C; Column temperature: 100 °C kept for 60 min. Carrier gas: N_2 (purity > 99.999%). Quantitative method: External standard method using the peak areas. Instrument model: Shimadzu 2014C; Sampling method: Off-line.

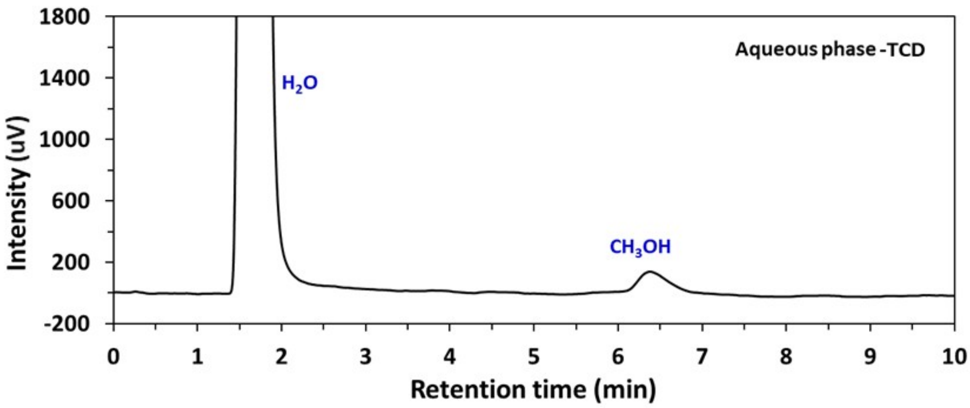


Figure S7. Typical GC-TCD spectrum of aqueous phase products.

Gas Chromatographic Conditions: Chromatographic Column: Porapak Q (2 m * 4 mm * 40 um) capillary column; Injection temperature: 220 °C; Detector temperature: 180 °C; Column temperature: 50 °C kept for 30 min; Carrier gas: H_2 (purity > 99.999%). Quantitative method: External standard method using the peak areas. Instrument model:

Shimadzu 2014C; Sampling method: Off-line.

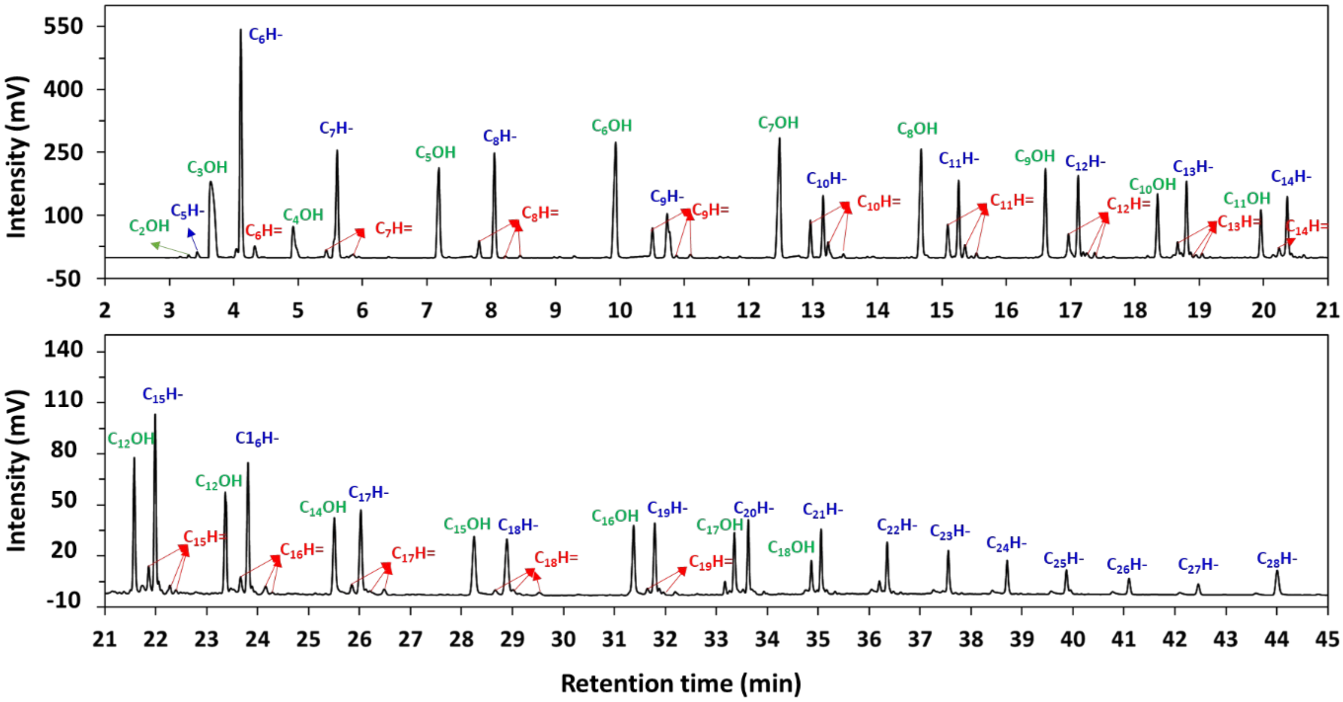


Figure S8. Typical GC-FID spectrum of oil phase products. C_xOH, $C_xH=$ and C_xH- represented alcohols, olefins, and paraffins containing x carbon atoms, respectively.

Gas Chromatographic Conditions: Chromatographic Column: WondaCap (30 m * 0.25 mm * 25 um) capillary column; Injection temperature: 300 °C; Detector temperature: 320 °C; Column temperature: 50 °C kept for 5 min, then rise up to 200 °C at a speed of 10 °C/min and kept for 10 min, finally rise up to 300 °C at a speed of 20 °C/min and kept for 10 min; Carrier gas: N_2 (purity > 99.999%). Quantitative method: Area normalization. Instrument model: Shimadzu 2014C; Sampling method: Off-line.

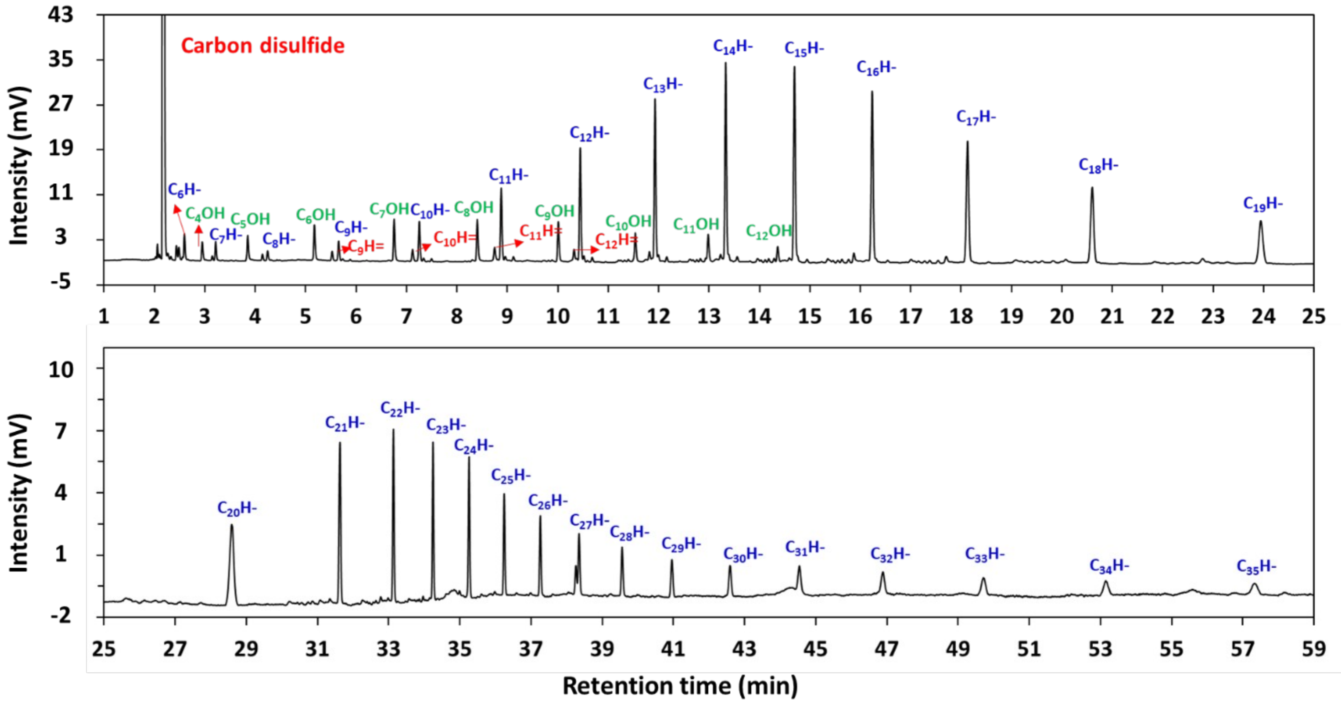


Figure S9. Typical GC-FID spectrum of wax phase products. C_xOH, C_xH= and C_xH- represented alcohols, olefins, and paraffins containing x carbon atoms, respectively.

Chromatographic Columns: Chromatographic column: WondaCap (30 m ∗ 0.25 mm ∗ 25 um) capillary column; Injection temperature: 320 °C; Detector temperature: 330 °C; Column temperature: 80 °C kept for 2 min, then rise up to 200 °C at a speed of 10 °C/min and kept for 15 min, finally rise up to 300 °C at a speed of 20 °C/min and kept for 40 min; Carrier gas: N_2 (purity > 99.999%). Quantitative method: Area normalization. Instrument model: Shimadzu 2014C; Sampling method: Off-line.

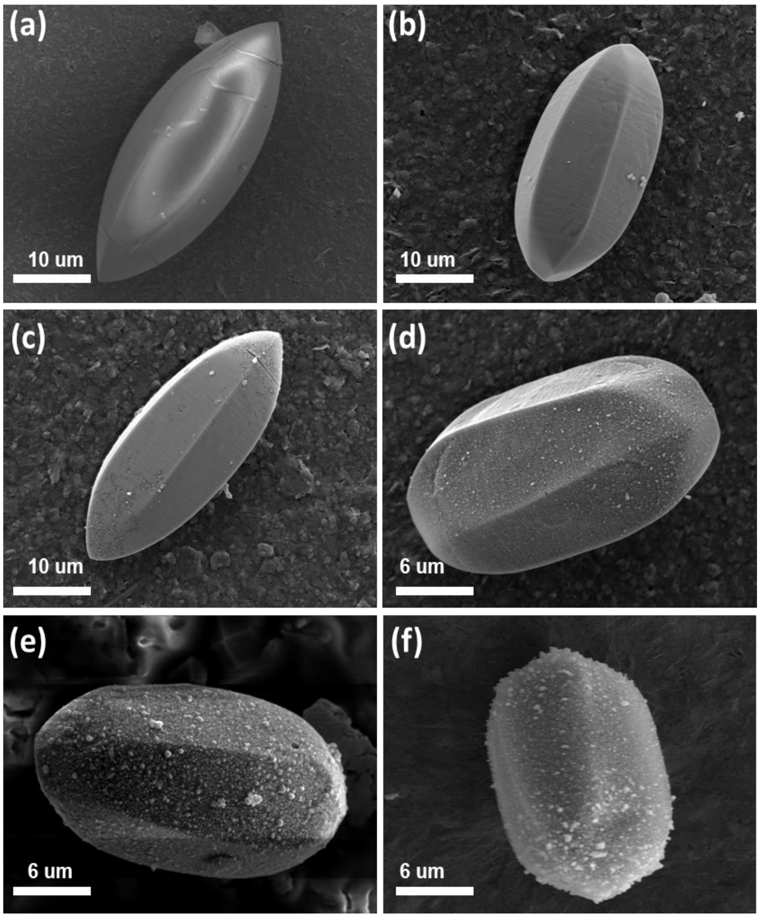


Figure S10. The low-magnification FESEM images of (a) CoMn-MOF-74, the derived (b) Co/MnO_x@quasi-MOF-74, (c) Co/MnO_x@C-500, (d) Co/MnO_x@C-600, (e) Co/MnO_x@C-700, and (f) Co/MnO_x@C-800.

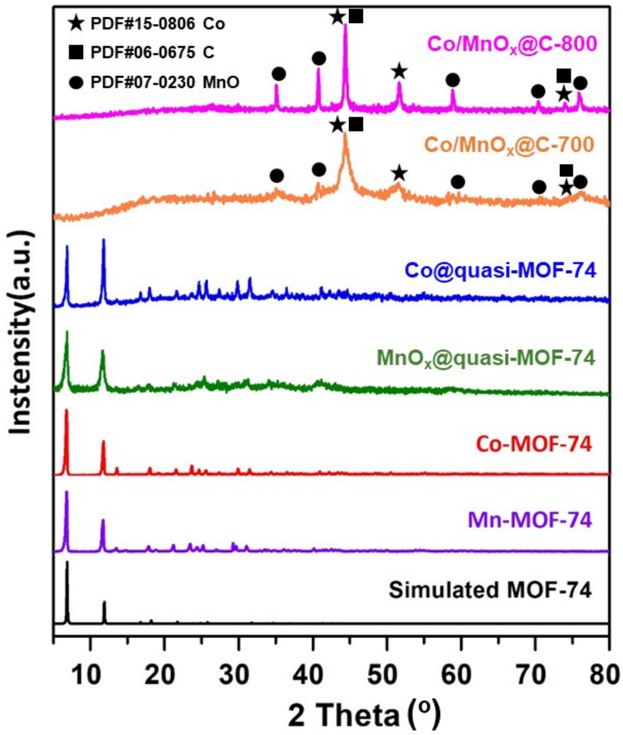


Figure S11. XRD patterns of Mn-MOF-74, Co-MOF-74, MnO_x@quasi-MOF-74, Co@quasi-MOF-74, Co/MnO_x@C-700, and Co/MnO_x@C-800.

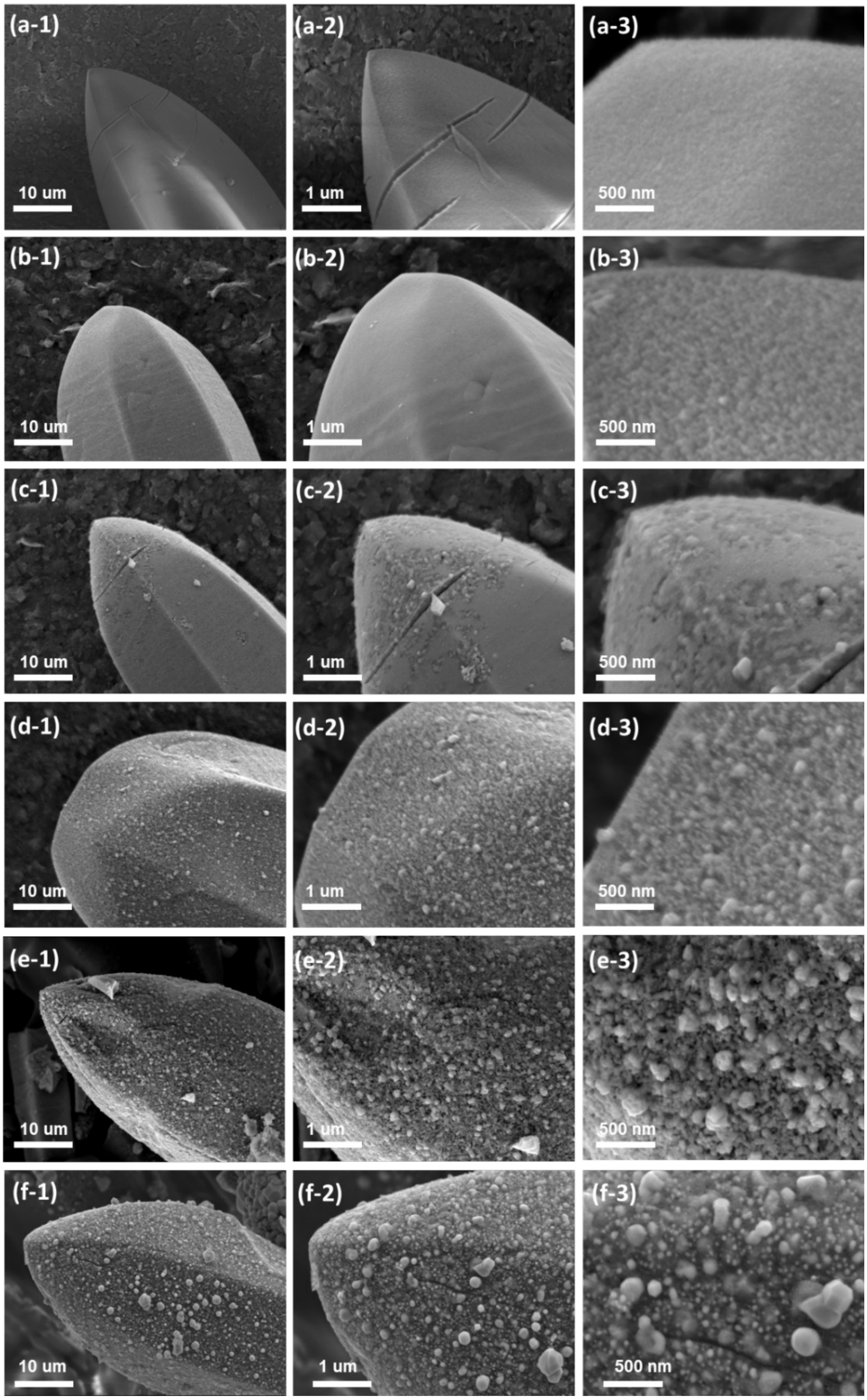


Figure S12. The high-magnification FESEM images of (a_{1-3}) CoMn-MOF-74, the derived (b_{1-3}) Co/MnO_x@quasi-MOF-74, (c_{1-3}) Co/MnO_x@C-500, (d_{1-3}) Co/MnO_x@C-600, (e_{1-3}) Co/MnO_x@C-700, and (f_{1-3}) Co/MnO_x@C-800.

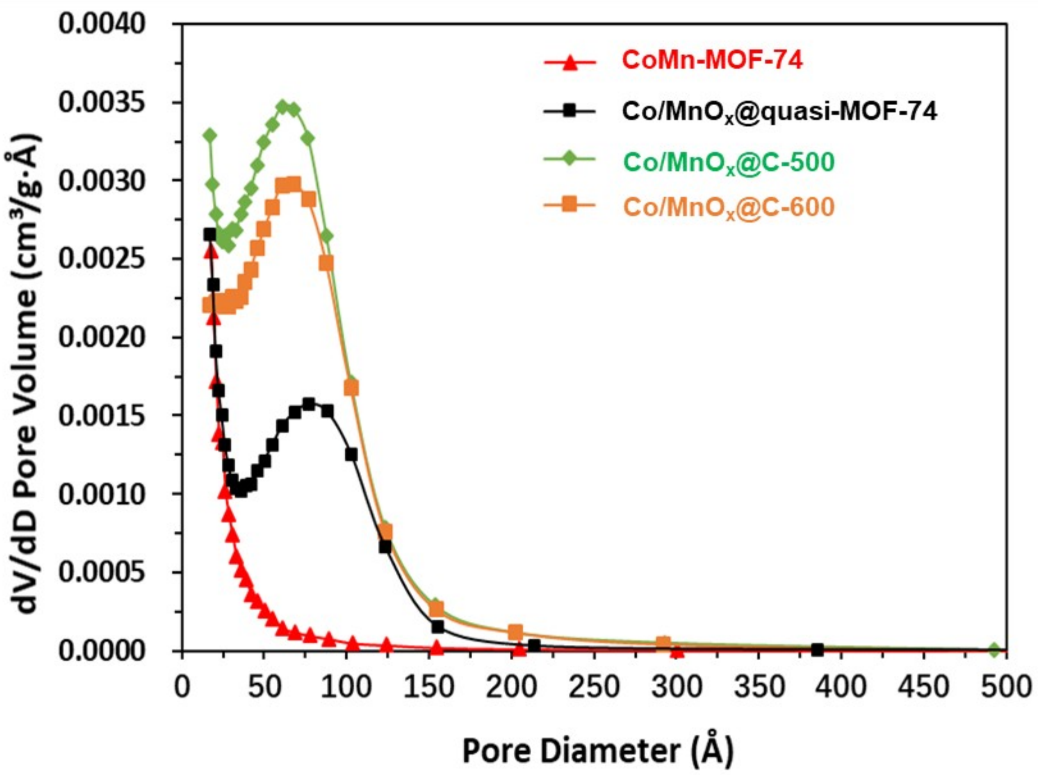


Figure S13. The pore diameter distribution of CoMn-MOF-74, Co/MnO_x@quasi-MOF-74 and Co/MnO_x@C-500, 600 catalysts.

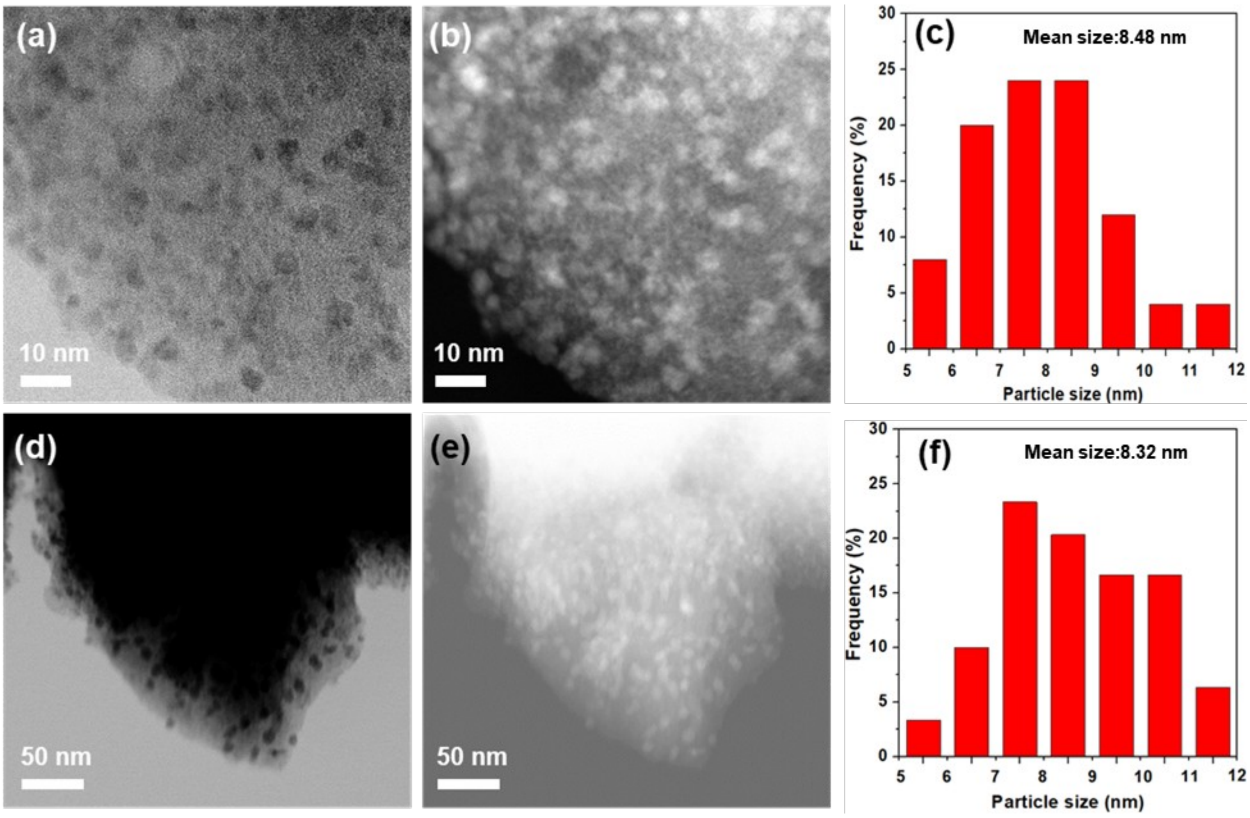


Figure S14. HAABF-STEM (a), HAADF-STEM (b), bright field (d) and dark field (e) TEM images of the Co/MnO_x@quasi-MOF-74 catalyst; Size distribution of nanoparticles (c) and (f) in (a) and (d), respectively.

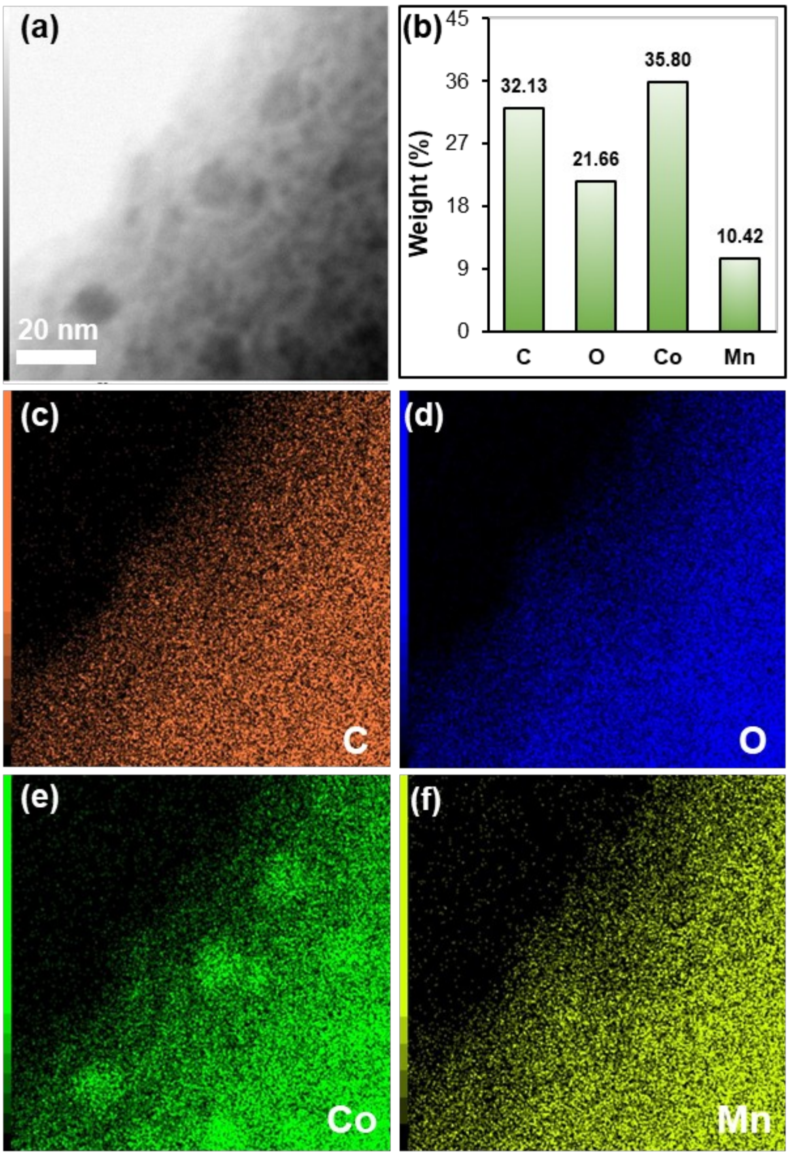


Figure S15. HRTEM image (a) and corresponding elements mapping (c, d, e and f), (b) Elemental analysis results of Co/MnO_x@quasi-MOF-74.

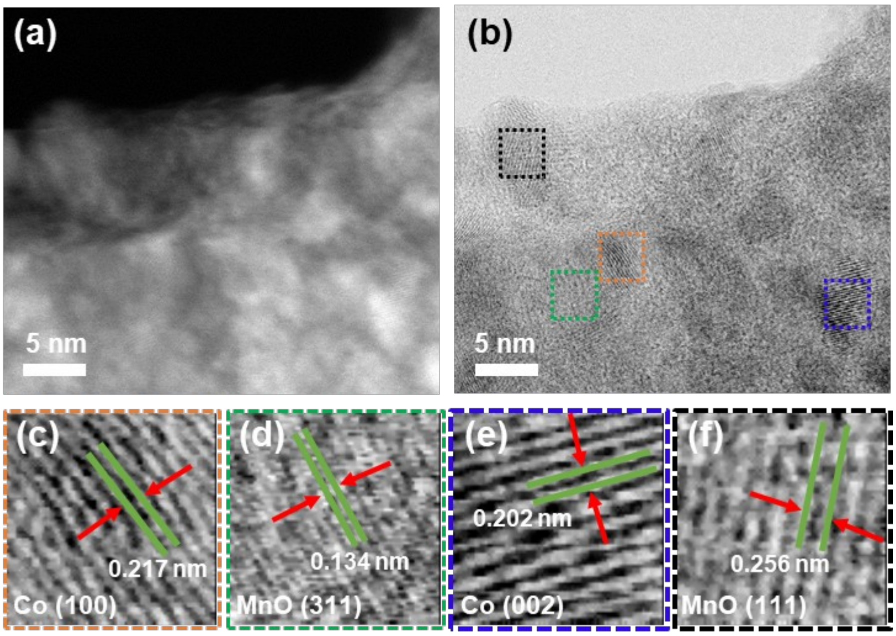


Figure S16. HAADF-STEM (a) and HAABF-STEM (b) images of Co/MnO_x@quasi-MOF-74. Enlarged high-resolution HAADF images taken from the selected areas indicated by (c) red, (d) green, (e) blue, and (f) black, rectangles in (b).

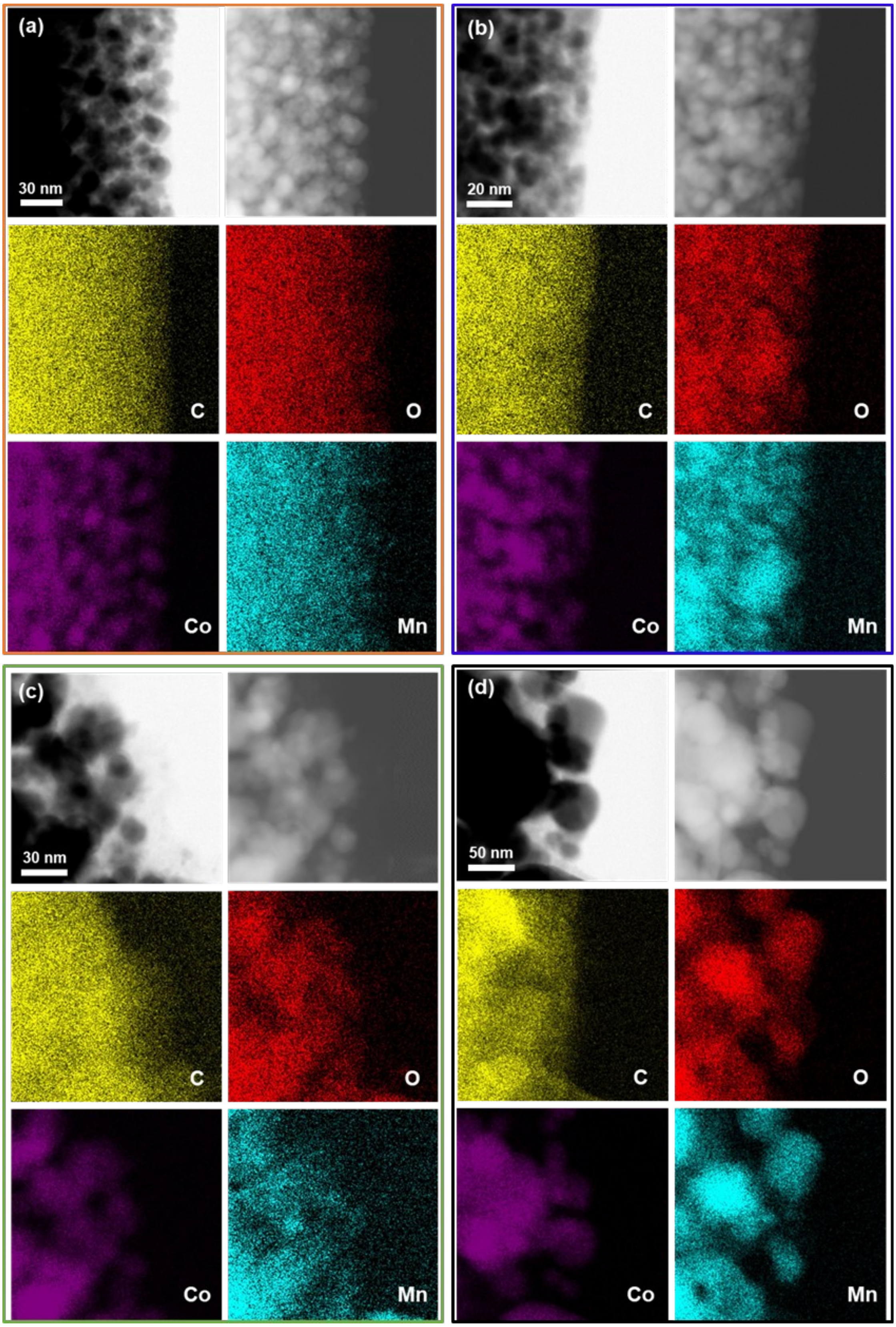


Figure S17. TEM and corresponding EDS elemental mapping images of (a) Co/MnO_x@C-500, (b) Co/MnO_x@C-600, (c) Co/MnO_x@C-700, and (d) Co/MnO_x@C-800.

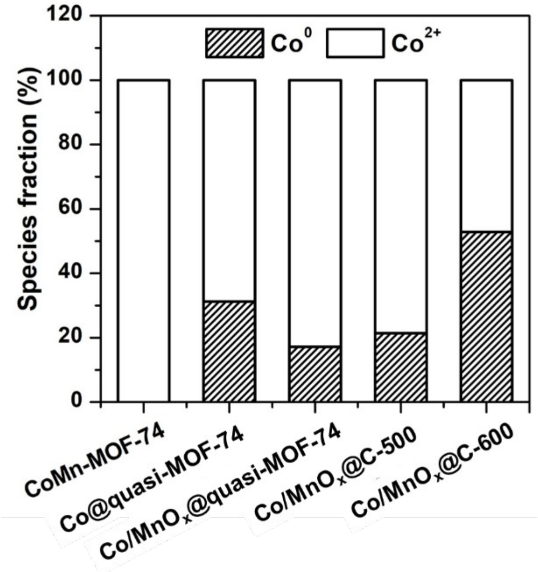


Figure S18. The results of deconvolution and quantitative evaluation of the Co 2p of CoMn-MOF-74, Co@quasi-MOF-74, Co/MnO$_x$@quasi-MOF-74, and Co/MnO$_x$@C-X catalysts.

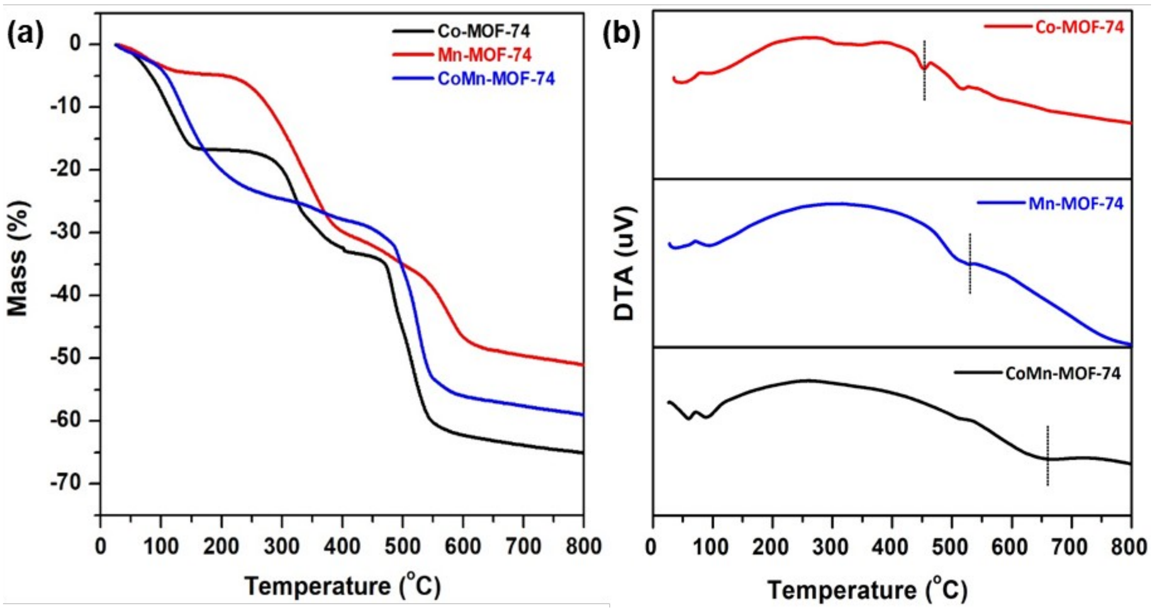


Figure S19. (a) TGA and (b) differential thermal analysis (DTA) plots of Co-MOF-74, Mn-MOF-74, and CoMn-MOF-74 measured from 30 to 800 °C in N_2 atmosphere at a heating rate of 5 °C·min^{-1}.

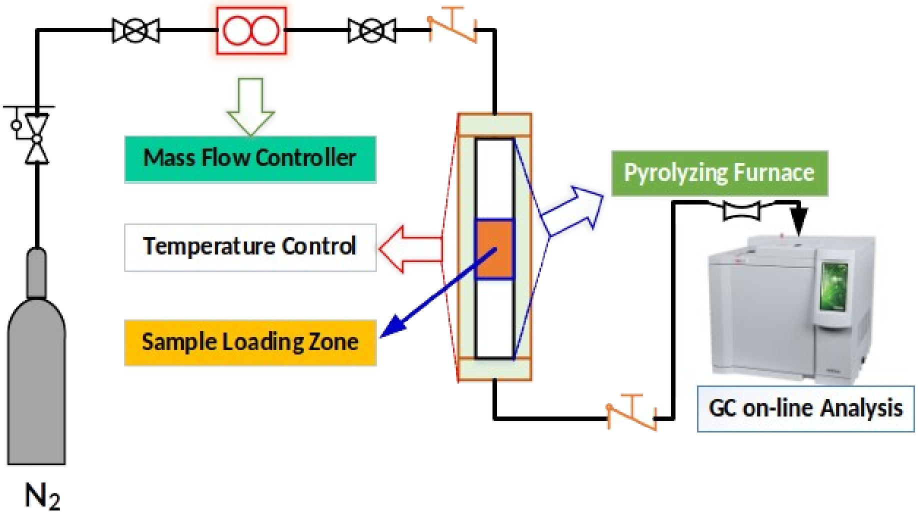


Figure S20. The schematic diagram of temperature-programmed pyrolysis (TPP) apparatus.

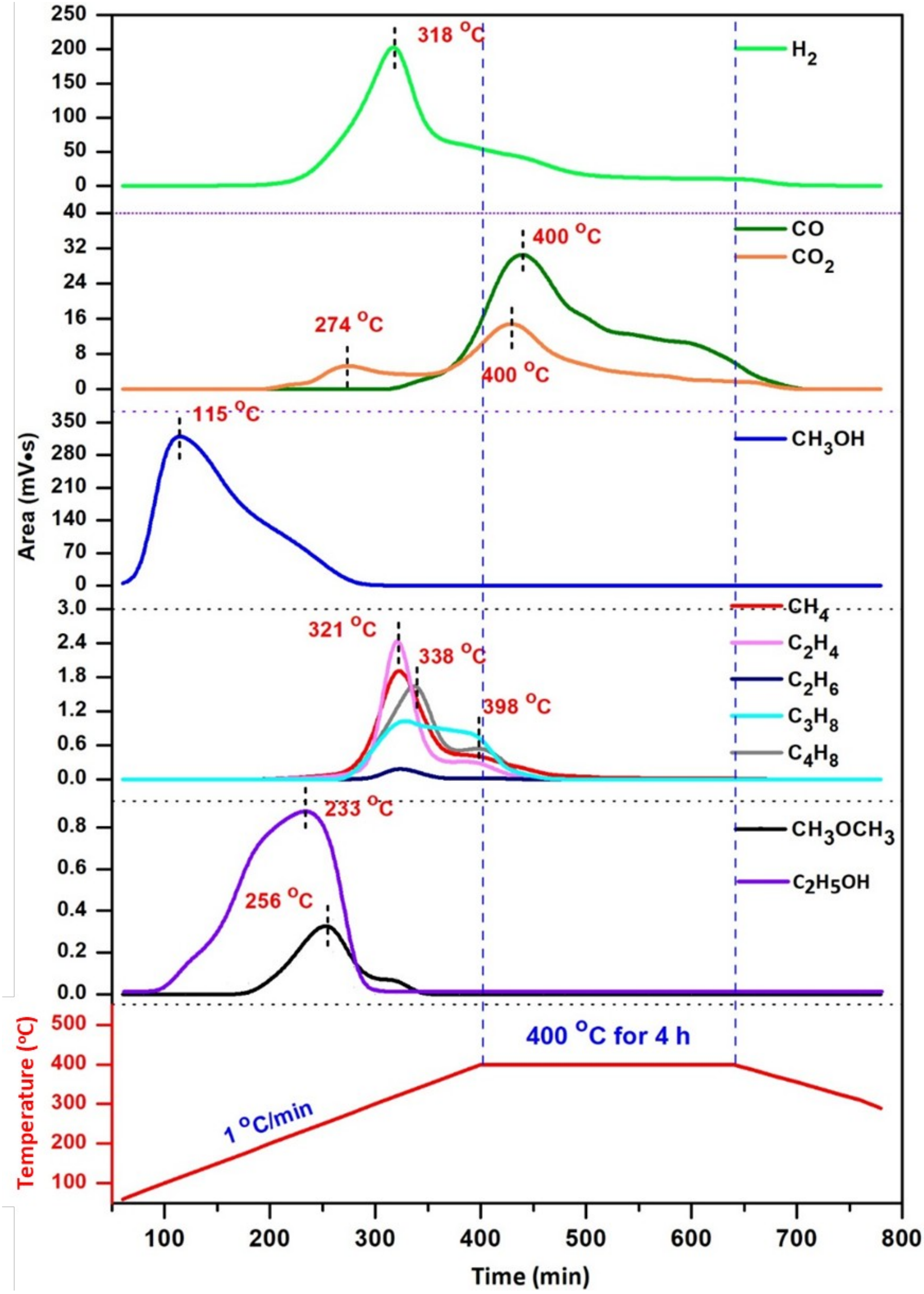


Figure S21. Plots for elucidating the produced gaseous species during the temperature programming pyrolysis of the CoMn-MOF-74.

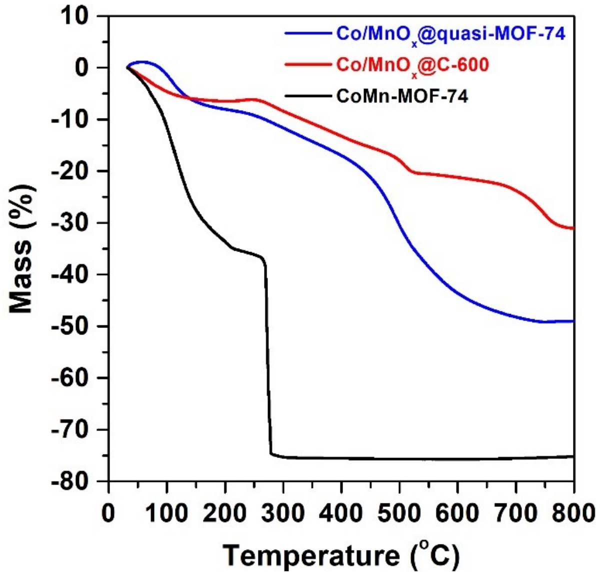


Figure S22. TGA plots of Co/MnO_x@quasi-MOF-74, Co/MnO_x@C-600, and CoMn-MOF-74 measured from 30 to 800 °C in air atmosphere at a heating rate of 10 °C·min^{-1}.

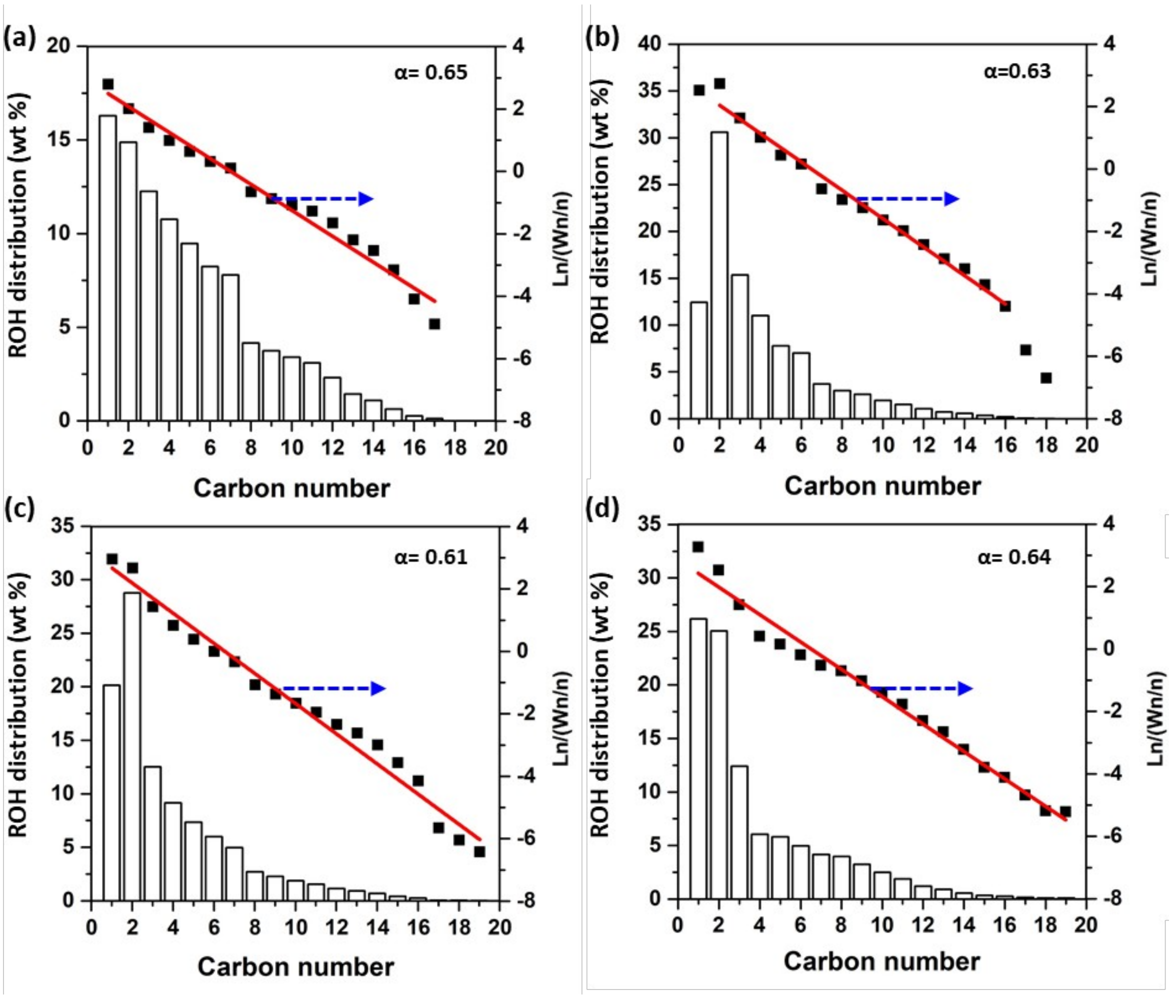


Figure S23. The ROH distribution and corresponding ASF plots of (a) Co@quasi-MOF-74, (b) Co/MnO_x@C-500, (c) Co/MnO_x@C-600, and (d) Co/MnO_x@C-700 catalysts. Reaction conditions: 230 ºC, 3.0 MPa, CO/H_2 = 1/2, GHSV=4500 mL·g^{-1}·h^{-1}. TOS = 24 h.

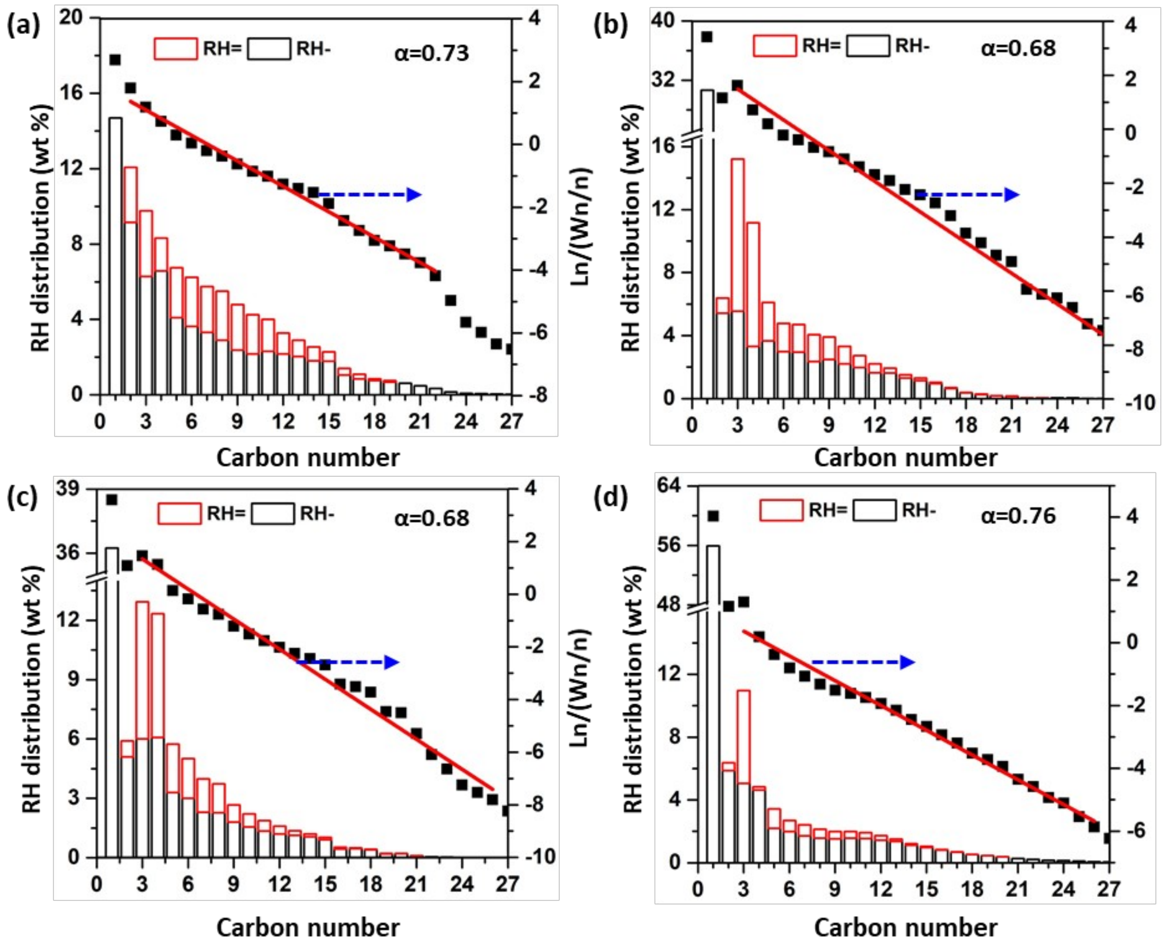


Figure S24. The RH distribution and corresponding ASF plots of (a) Co@quasi-MOF-74, (b) Co/MnO_x@C-500, (c) Co/MnO_x@C-600, and (d) Co/MnO_x@C-700 catalysts. Reaction conditions: 230 ºC, 3.0 MPa, CO/H_2 = 1/2, GHSV=4500 mL·g^{-1}·h^{-1}.TOS = 24 h.

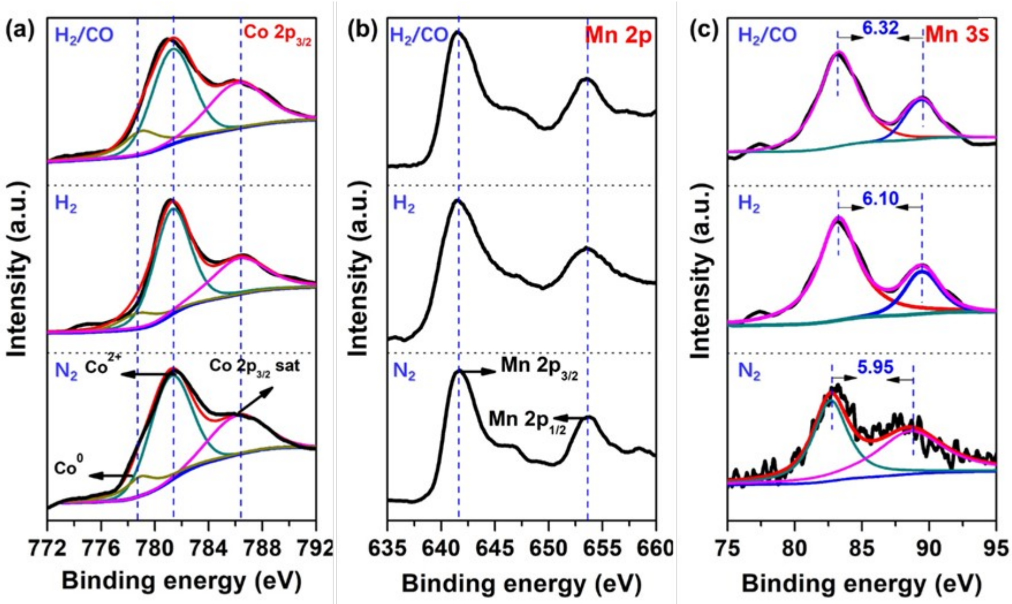


Figure S25. Semi-in situ XPS analysis of Co/MnO_x@quasi-MOF-74. The catalyst was dried at 250 ºC in N_2 for 1.0 h (labeled as N_2), then pretreated with a H_2/N_2 of 30/70 (v/v) at 300 ºC for 2.0 h (labeled as H_2), and finally treated with syngas (H_2/CO = 2) at 230 ºC for 2.0 h (labeled as H_2/CO). The whole process was not exposed to air.

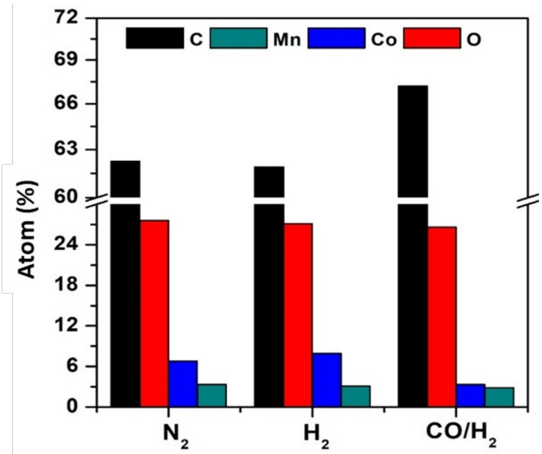


Figure S26. Elemental composition of the surface of Co/MnO_x@quasi-MOF-74 catalyst observed from semi-in situ survey XPS spectrum. The catalyst was dried at 250 ºC in N_2 for 1.0 h, then pretreated with a H_2/N_2 of 30/70 (v/v) at 300 ºC for 2.0 h, and finally treated with syngas (H_2/CO = 2) at 230 ºC for 2.0 h. The whole process was not exposed to air.

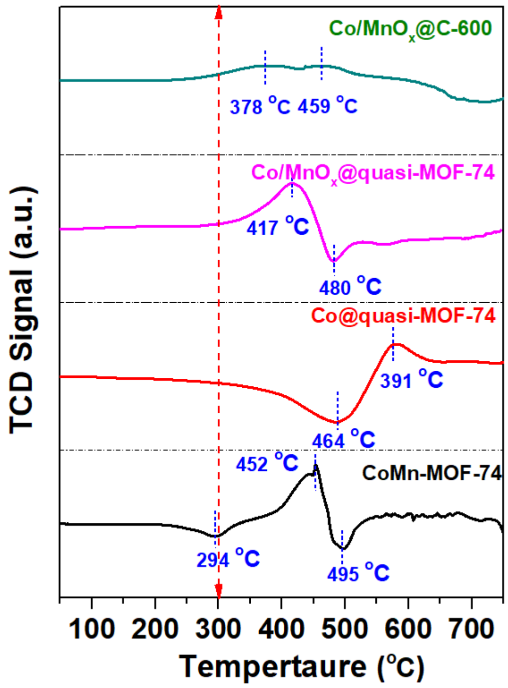


Figure S27. H_2-TPR plots for Co@quasi-MOF-74, and CoMn-MOF-74 and its derived catalysts.

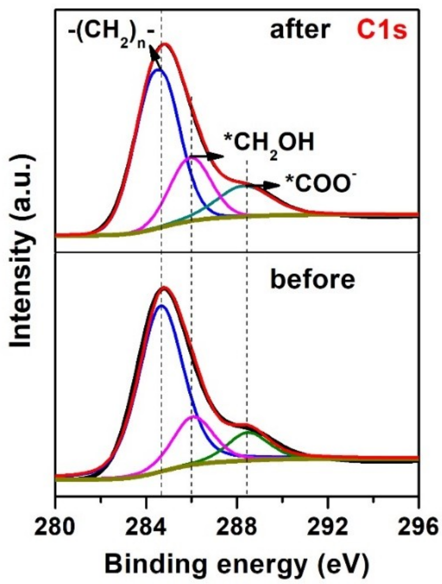


Figure S28. C1s XPS spectra of Co/MnO_x@quasi-MOF-74 catalyst before and after 2.0 h of reaction.

Note: The most intense line at 284.8 eV is associated with an elementary carbon intercalated into the bulk of catalyst and/or with hydrocarbon residues adsorbed on the surface. Two weak peaks at 286.2 and 289.6 eV are assigned to the carbon atoms bonded with several oxygen atoms (e.g. carboxyl or carbonate groups) [1].

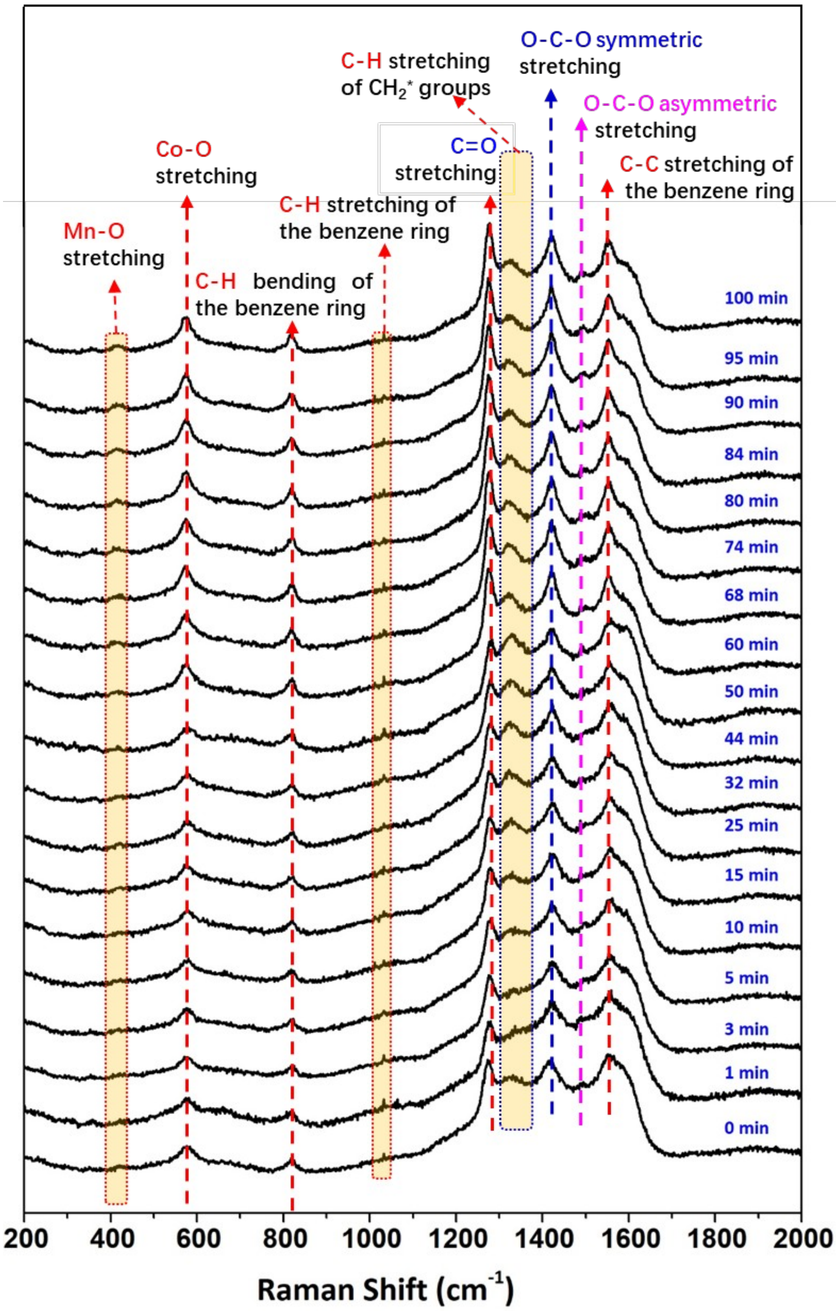


Figure S29. In situ Raman spectra of CO hydrogenation reaction on the pretreated Co/MnO_x@quasi-MOF-74 catalyst at 230 ºC, 0.1 MPa, 30 mL·min^{-1} of H_2/CO = 2 mixture.

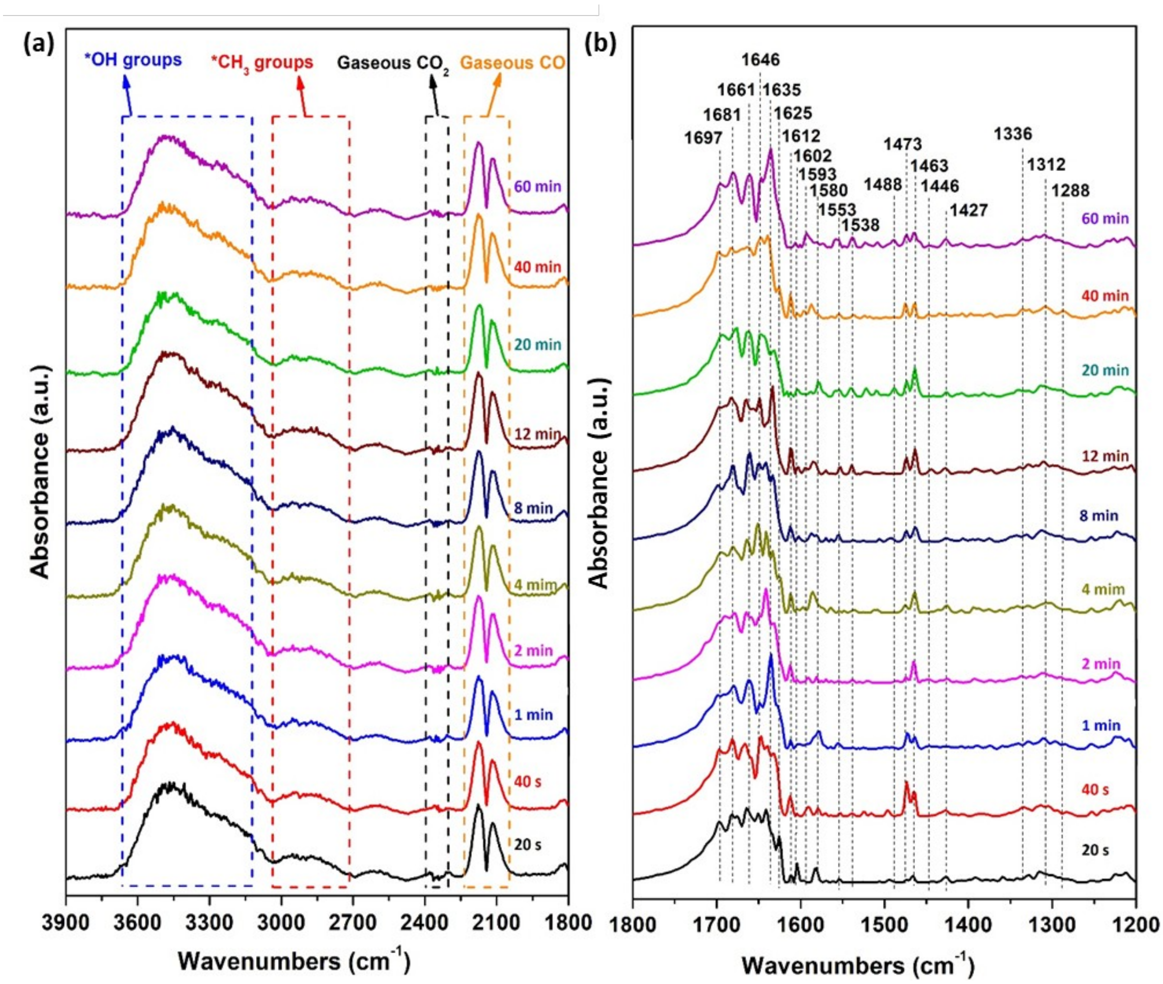


Figure S30. In situ DRIFT spectra of the CO hydrogenation reaction on the pretreated Co/MnO_x@C-600 catalyst in the regions (a) 3900-1800 cm^{-1} and (b) 1800-1200 cm^{-1} (Reaction conditions: 10 mL·min^{-1} CO + 20 mL·min^{-1} H_2, 0.1 MPa, 230 °C).

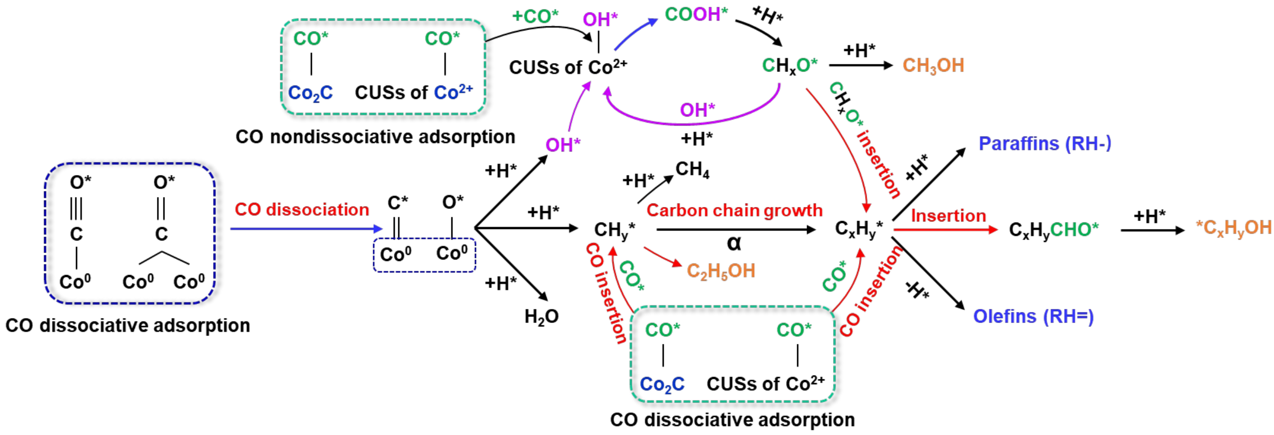


Figure S31. Proposed mechanism of ROH syntheses from CO hydrogenation over Co/MnO$_x$@quasi-MOF-74 catalyst.

S3. Supplementary Tables

Table S1 N_2 sorption measurements of catalysts.

	CoMn-MOF-74	Co/MnO_x@quasi-MOF-74	Co/MnO_x@C-500	Co/MnO_x@C-600
$S_{BET}{}^{a}$ ($m^2·g^{-1}$)	640.48	552.71	273.74	202.56
$S_{meso}{}^{b}$ ($m^2·g^{-1}$)	73.82	132.78	202.69	242.55
$S_{micro}{}^{c}$ ($m^2·g^{-1}$)	566.66	419.93	71.05	40.01
S_{meso}/S_{micro}	**0.13**	**0.32**	**2.85**	**12.12**
$D_{average}{}^{d}$ (Å)	30.81	55.16	55.29	48.82
$V_{micro}{}^{e}$ ($cm^3·g^{-1}$)	0.30	0.18	0.02	0.015
$V_{meso}{}^{f}$ ($cm^3·g^{-1}$)	0.04	0.11	0.23	0.19
V_{meso}/V_{micro}	**0.13**	**0.61**	**11.50**	**19.30**

a: Specific surface area calculated by the BET method; b: Mesoporous surface area evaluated by the BJH method from the adsorption branches of isotherms. c: Microporous surface area determined by $S_{BET}-S_{meso}$. d: Average pore diameter evaluated by the BJH method. e: V_{micro}: micropore volume determined by t-plot method. f: V_{meso}: mesopore volume determined by $V_{total}-V_{micro}$.

Table S2 The overall composition of the CoMn-MOF-74, Co/MnO_x@quasi-MOF-74 and Co/MnO_x@C-X catalysts.

Catalyst	Co (wt%)	Mn (wt%)	C (wt%)	H (wt%)	O (wt%)	Co/Mn	C/O
CoMn-MOF-74	25.52	7.47	30.85	4.18	31.98	3.42	0.96
Co/MnO_x@quasi-MOF-74	32.12	10.52	28.94	3.86	24.56	3.05	1.18
Co/MnO_x@C-500	40.72	13.91	24.99	0.96	19.43	2.93	1.28
Co/MnO_x@C-600	49.50	16.14	20.56	0.62	13.18	3.07	1.56

Note: The contents of Co and Mn elements were detected by ICP-OES, C and H elements were detected by elemental analyzer, while O element in the table was calculated by the subtraction method in the percentage of weight.

Table S3 The surface composition of the CoMn-MOF-74 Co/MnO$_x$@quasi-MOF-74 and Co/MnO$_x$@C-X catalysts detected by XPS.

Catalyst	Co (wt%)	Mn (wt%)	C (wt%)	O (wt%)	Co/Mn	C/O
CoMn-MOF-74	38.13	23.44	19.68	18.75	1.63	1.05
Co/MnO$_x$@quasi-MOF-74	25.71	15.36	33.93	25.00	1.67	1.36
Co/MnO$_x$@C-500	22.18	14.74	47.99	15.09	1.50	3.18
Co/MnO$_x$@C-600	18.85	13.49	55.93	11.76	1.34	4.76

Table S4 Effect of GHSV on the CO hydrogenation using Co/MnO_x@quasi-MOF-74.

GHSV/ ($mL \cdot g \cdot h^{-1}$)	Conv. (%)	Product distribution (wt%)							$ROH+RH^{=}$	$C_{2+}OH/ROH$	CTY/ ($mg_{CO} \cdot g_{cat} \cdot h^{-1}$)	STY/ ($mg_{ROH+RH} \cdot g_{cat} \cdot h^{-1}$)	α_{ROH}	α_{RH}
		CH_4	$C_{2+}H$	$RH^{=}$	CH_3OH	$C_{2+}OH$	ROH	CO_2						
15000	10.5	8.2	34.1	20.1	3.8	33.9	37.7	0.0	57.7	90.0	619.9	206.0	0.57	0.65
12000	11.1	8.9	34.9	21.2	5.0	29.7	34.7	0.0	55.9	85.6	525.2	166.8	0.60	0.69
7500	14.9	10.1	29.4	24.7	4.4	31.4	35.8	0.0	60.5	87.6	439.4	152.2	0.63	0.70
4500	21.4	10.0	28.6	21.6	3.0	35.9	39.0	0.8	60.5	92.2	379.7	91.3	0.64	0.72

Other conditions: P = 3.0 MPa, H_2/CO = 2, and T = 230 ºC. TOS = 24 h.

Table S5 Effect of reaction temperature on the CO hydrogenation using Co/MnO_x@quasi-MOF-74.

Temp./ (^oC)	Conv. (%)	Product distribution (wt%)							$ROH+RH^=$	$C_{2+}OH/ROH$	CTY/ ($mg_{CO}\cdot g_{cat}\cdot h^{-1}$)	STY/ ($mg_{ROH+RH}\cdot g_{cat}\cdot h^{-1}$)	α_{ROH}	α_{RH}
		CH_4	$C_{2+}H$	$RH^=$	CH_3OH	$C_{2+}OH$	ROH	CO_2						
200	6.7	3.8	30.9	20.4	3.4	45.4	48.7	0.0	69.1	93.1	117.8	47.8	0.62	0.70
220	12.1	7.0	29.3	20.6	3.2	39.3	42.5	0.0	64.1	92.5	214.2	74.0	0.63	0.71
230	21.4	10.0	28.6	21.6	3.0	35.9	39.0	0.8	60.5	92.2	383.2	91.3	0.64	0.72
250	38.7	18.2	33.1	14.3	1.1	25.2	26.3	8.1	40.6	93.8	686.4	136.0	0.66	0.78

Other conditions: P= 3.0 MPa, $H_2/CO = 2$, and GHSV = 4500 $mL\cdot g^{-1}\cdot h^{-1}$. TOS = 24 h.

Table S6 Effect of GHSV on the CO hydrogenation using Co/MnO_x@C-600 catalyst.

GHSV/ ($mL \cdot g \cdot h^{-1}$)	Conv. (%)	Product distribution (wt%)							ROH+RH=	$C_{2+}OH$/ROH	CTY/ ($mg_{CO} \cdot g_{cat} \cdot h^{-1}$)	STY/ ($mg_{ROH+RH} \cdot g_{cat} \cdot h^{-1}$)	α_{ROH}	α_{RH}
		CH_4	$C_{2+}H$	RH=	CH_3OH	$C_{2+}OH$	ROH	CO_2						
4500	49.9	29.8	32.7	18.9	2.8	10.7	13.5	5.1	32.4	79.2	88.4	163.9	0.62	0.67
7500	23.3	27.1	33.9	16.8	3.3	12.2	15.5	6.8	32.3	78.6	68.7	128.5	0.68	0.70
12000	13.9	23.5	35.4	16.1	5.1	17.6	22.8	2.3	38.9	77.5	65.6	145.2	0.70	0.73
15000	10.7	21.0	34.0	16.0	6.1	21.6	27.7	1.4	43.7	78.0	62.9	157.1	0.73	0.74

Other conditions: P = 3.0 MPa, H_2/CO = 2, and T = 230 ºC. TOS = 24 h.

Table S7 Catalytic performance of state of the art catalysts in the selective conversion of syngas to $C_{2+}OH$.

Catalyst	T (ºC)	P (MPa)	GHSV ($mL \cdot g_{cat}^{-1} \cdot h^{-1}$)	H_2/CO	X_{CO} (%)	CH_4 (%)	S_{ROH} (%)	$C_{2+}OH/ROH$ (%)	S_{CO2} (%)	Ref.
Co/MnO_x@quasi-MOF-74	200	3.0	4500	2	6.65	3.4	45.4	93.1	0.0	This work
Co/MnO_x@quasi-MOF-74	220	3.0	4500	2	12.1	7.0	42.4	92.5	0.0	This work
Co/MnO_x@quasi-MOF-74	230	3.0	4500	2	21.4	9.98	39.0	92.1	0.8	This work
CoMn|CuZnAlZr[(a)]	220	6.0	2000	2	12.4	4.4	54.5	93.1	6.3	[2]
Cu_4Fe_1	260	1.0	2400	2	53.2	3.7	29.8	49.1	30.0	[3]
Co_1/CeO_2	260	5.0	N/A	2	1.0	23.8	21.7	73.4	0.8	[4]
CuCoAl	270	2.5	7500	2	36.2	24.0	40.7	54.9	0.5	[5]
Cs_2O-Cu/ZnO/Al_2O_3	320	5.4	3750	2	12.0	N/A	23.3	N/A	20	[6]
Co−Co_2C/AC	200	3.0	3665	2	25.3	N/A	37.0	87	4.6	[7]
Co1Mn/AC	220	3.0	2000	2	29.1	8.1	21.4	N/A	2.4	[8]
LKCG-0.1	290	4.0	6000	2	13.2	30.9	43.6	69.1	5.8	[9]
Rh/MnO/SiO_2	250	2.0	N/A	2	N/A	26.0	N/A	56.1	N/A	[10]
Co0.5Mn1La/AC	220	3.0	2000	2	18.3	N/A	26.8	N/A	2.5	[11]
CoMn-0.6Na	240	2.0	6000	2	11.6	18.2	19.8	N/A	31.6	[12]
3Cu−5Co/(Mn−Al)	260	5.0	5000	2	33.4	N/A	39.7	57.3	14.1	[13]
15Co3Fe/AC	220	3.0	4000	3	14.8	N/A	22.3	N/A	1.7	[14]
10K-Co_3O_4	270	4.0	N/A	N/A	0.17	10.9	16.0	N/A	N/A	[15]
Co_8P/SiO_2	280	5.0	5000	2	6.7	38.8	39.6	87.6	4.8	[16]
Cu@(CuCo-alloy)/Al_2O_3	220	2.0	2000	2	32.9	N/A	36.2	58.3	7.2	[17]
Co–Cu	270	2.0	18000	2	1.0	17.2	18.0	63.3	48.8	[18]
Cu@Co_3O_4	270	1.0	18000	2	N/A	9.8	20.9	23.4	59.6	[19]
$(Cu_1Co_2)_2$Al-LDHs	250	3.0	3900	2	53.6	N/A	39.2	91.0	2.2	[20]
2Co–2Pd/SiO_2	230	1.0	24000	2	0.05	11.7	20.2	22.27	N/A	[21]
K-Co-β-Mo_2C	300	8.0	2000	1	51.0	N/A	27.4	N/A	N/A	[22]
K-Co-MoS_2/AC	330	5.0	4800	2	14.3	N/A	26.7	81.3	N/A	[23]
K-Co-MoS_2/CNTs	350	5.0	3600	1	21.6	N/A	45.9	34.8	28.3	[24]
K-Co-MoP/SiO_2	275	5.0	3600	1	11.2	12.5	30.7	93.3	N/A	[25]
15Co5Fe/AC	220	3.0	4000	2	16.4	N/A	20.6	87.2	1.6	[26]

(a) Calculated on a CO_2-free basis.

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